

Istahadul Hoque, MATH-2407 (MRK 61R)

4BM

MATH-2407 [FINAL]

FOURIER SERIES

Signal: A signal is defined as any physical phenomenon that carries information and varies with time.

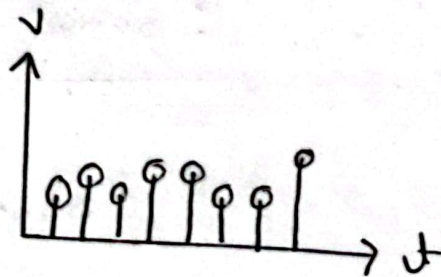
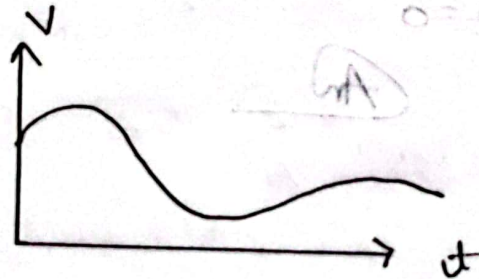
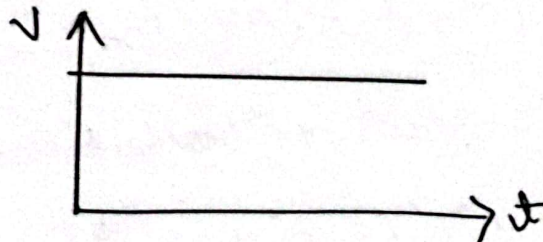


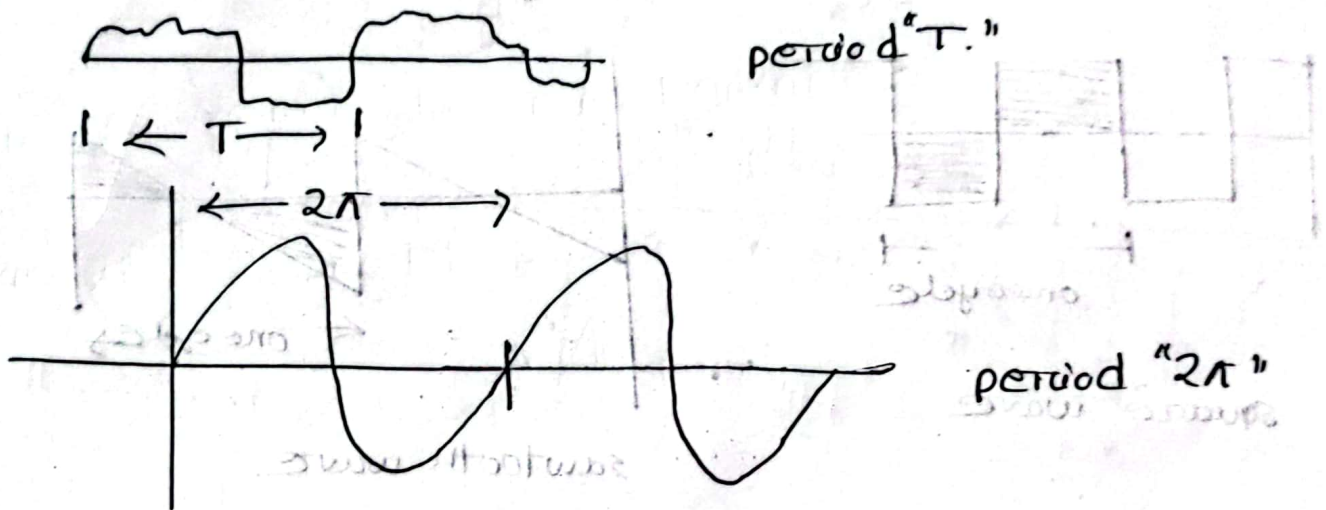
fig:2 Continuous vs discrete signal.

DC signal:

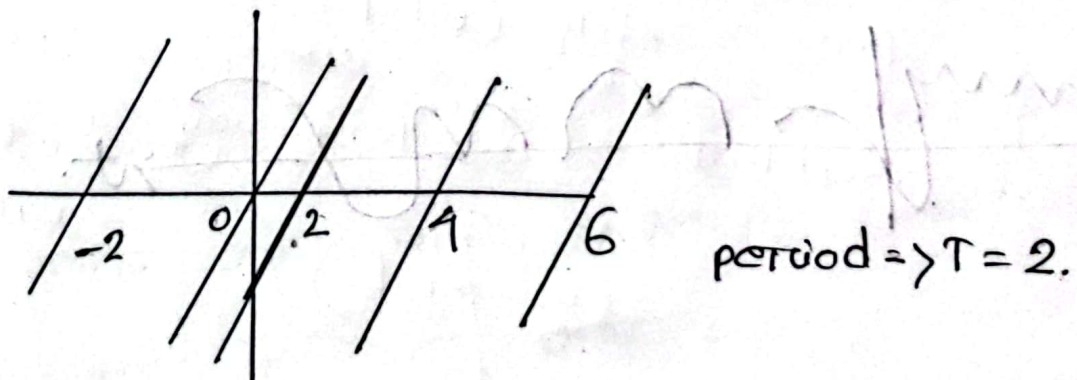


Periodic signal: A signal which is repeating itself

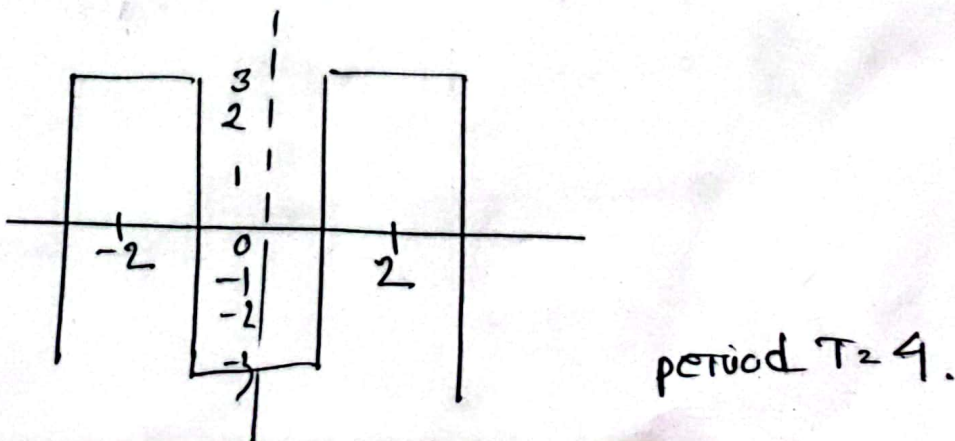
periodic signal.

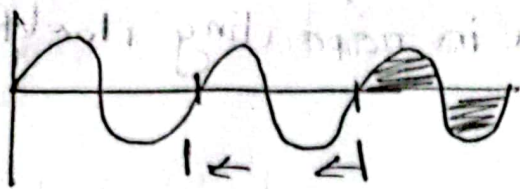


Ex-4 $f(t) = 3t$ $-1 \leq t \leq 1$
 $f(t) = f(t+2)$



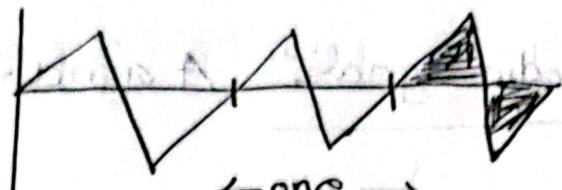
Ex-6





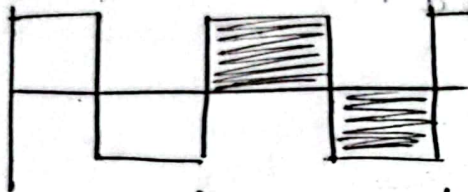
one cycle

sine wave



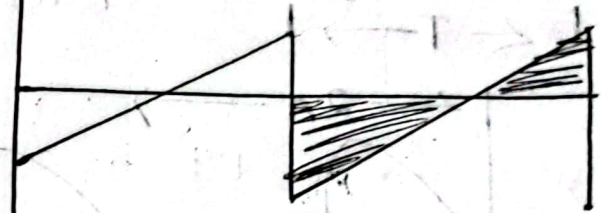
← one cycle →

triangular wave



one cycle

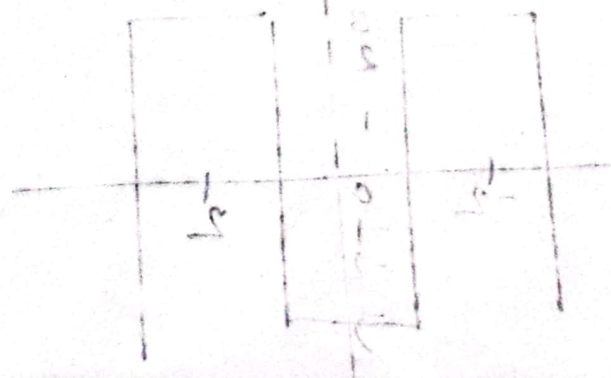
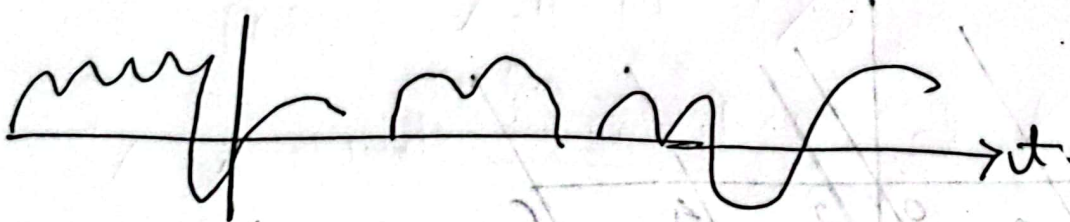
square wave

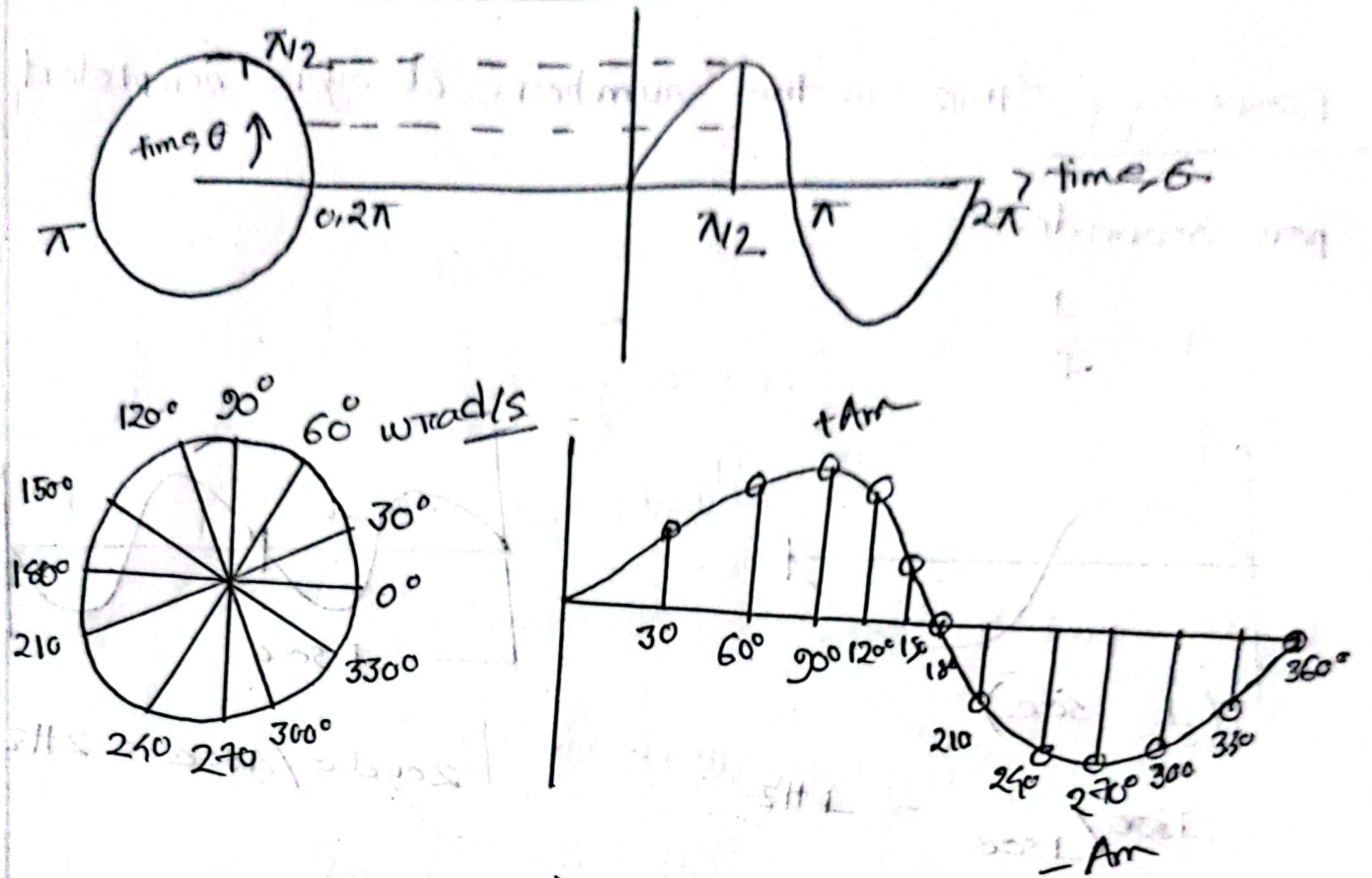


← one cycle →

sawtooth wave

Non-periodic wave:





$$A(t) = A_m \sin(\omega t + \phi)$$

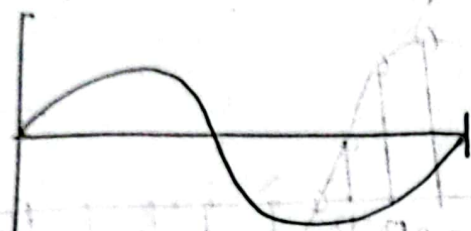
ϕ = phase difference A_m = max magnitude

এখানে চাকাটি $\omega \text{ rad/s}$ বেগে ঘুরছে; এখন 30° কোণ অতিক্রম
 ও গুরুত্ব করলে; আমরা জানি $\sin 30^\circ = \frac{1}{2}$ এবং তা জানা করে
 বসিয়ে দিলে, এখন $\sin 60^\circ = \frac{\sqrt{3}}{2}$ এবং $\sin 90^\circ = 1$, যা সবচেয়ে
 বড় (Am দ্বারা চিহ্নিত), $\sin 120^\circ = \frac{\sqrt{3}}{2}$, $\sin 150^\circ = \frac{1}{2}$ এবং
 বাকিগুলো বের করে জানা করে বসালে অর্থাৎ
 বিন্দুগুলো তুলে করলে আমরা $\sin$ দেখতে পাবো,

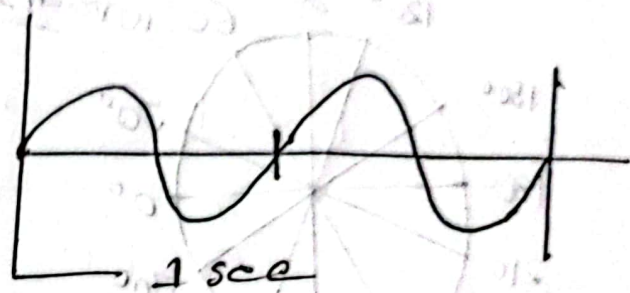
frequency: This is the number of cycle completed

per second

$$\therefore f = \frac{1}{T}$$



$$1 \text{ cycle} / 1 \text{ sec} = 1 \text{ Hz}$$



$$2 \text{ cycle} / 1 \text{ sec} = 2 \text{ Hz}$$

Expression of sinusoidal signal:

$$f(t) = A \sin(\omega t + \phi)$$

A = Amplitude

ω = frequency

ϕ = phase shift
with π to time t .

$$(\omega = 2\pi f)$$

Ex-10

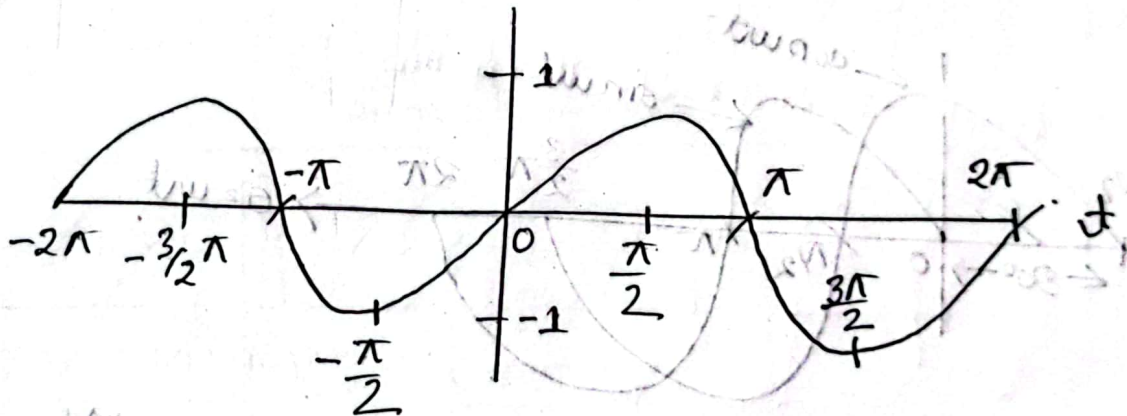
$$f(t) = A \sin(\omega t + \phi)$$

gf $A=1$, $\phi=0$, draw the graph of $f(t)=?$

$$\text{Then } f(t) = A \sin(\omega t + \phi)$$

$$= 1 \sin(\omega t + 0)$$

$$= \sin \omega t.$$



Ex-11

$$f(t) = A \sin(\omega t + \phi)$$

gf $A=1$, $\phi=90^\circ$, then draw the graph of

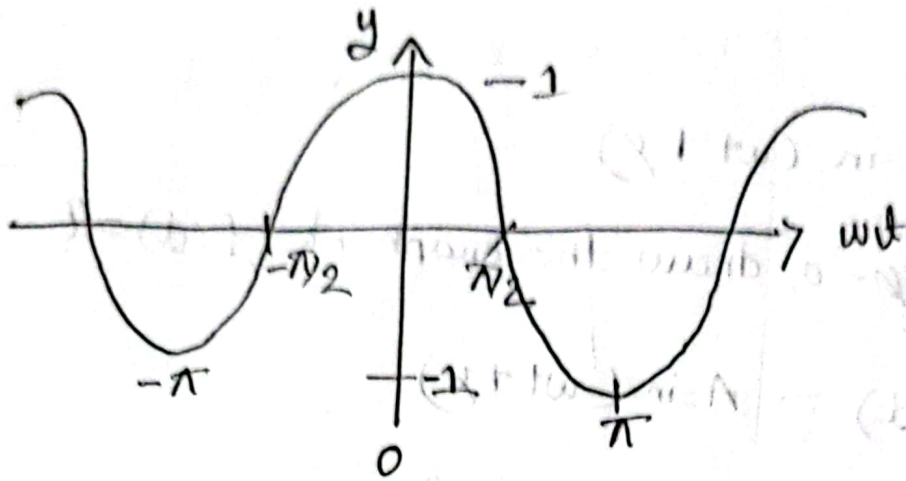
$$f(t) = ?$$

$$f(t) = A \sin(\omega t + \phi)$$

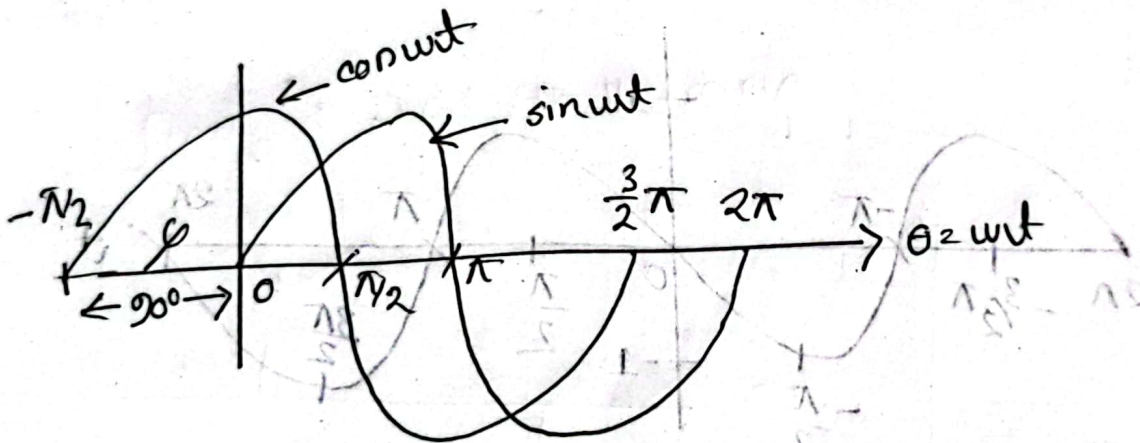
$$= 1 \sin(\omega t + 90^\circ)$$

$$= \sin(90^\circ + \omega t)$$

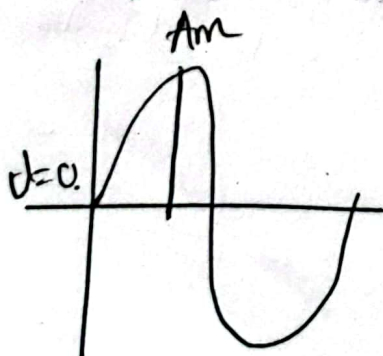
$$= \cos \omega t.$$



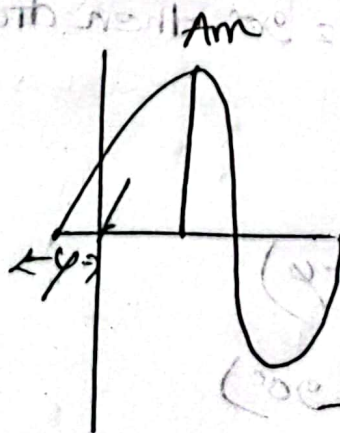
Ex-12 prob-4:



sine and cosine (wave) with phase diff. 90° .

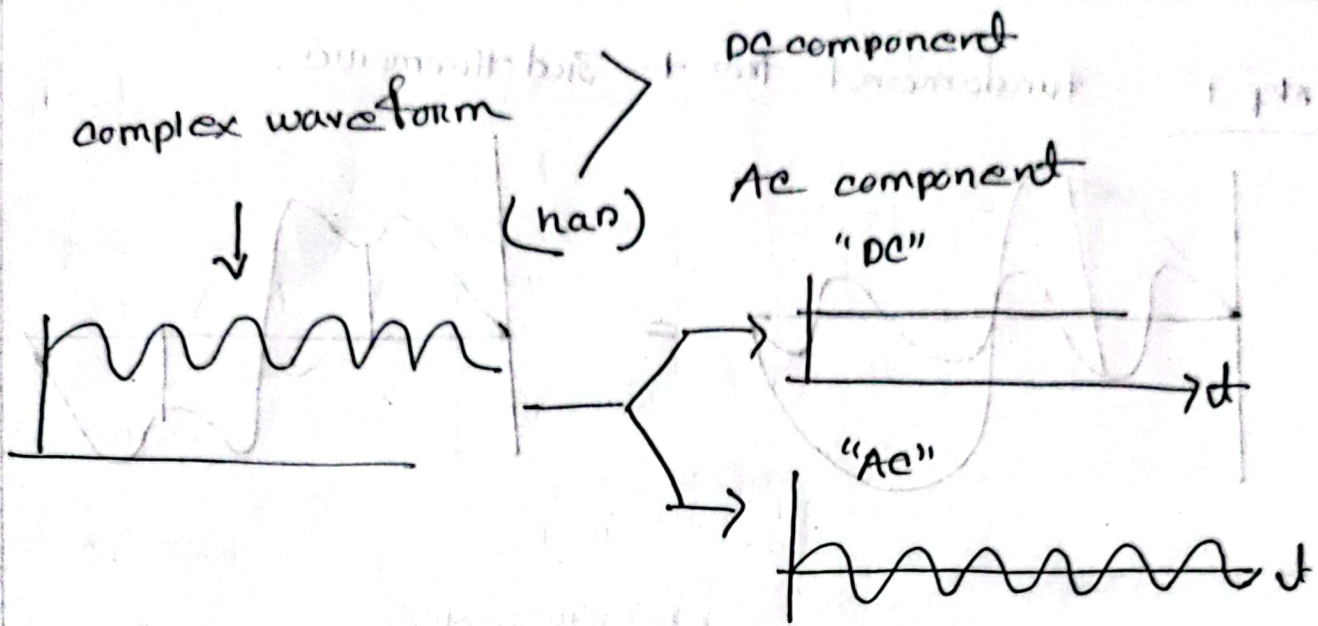


$$= A_m \sin wt$$

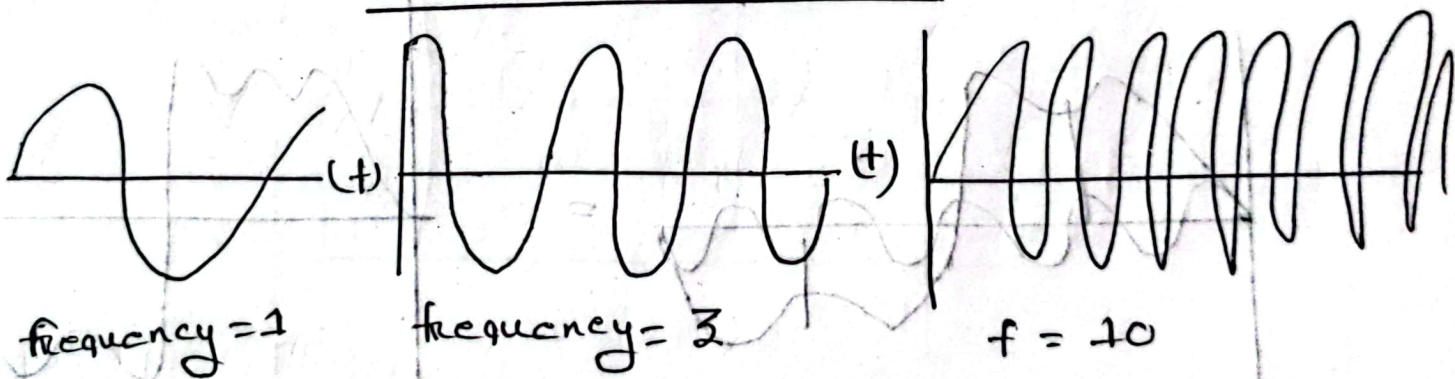


$$= A_m \sin (wt + \phi) = A_m \sin (wt + \phi)$$

two nos =



Sum of waves

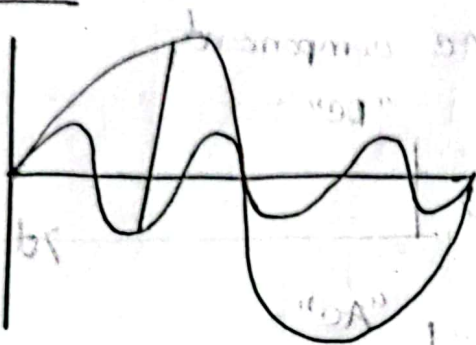


=

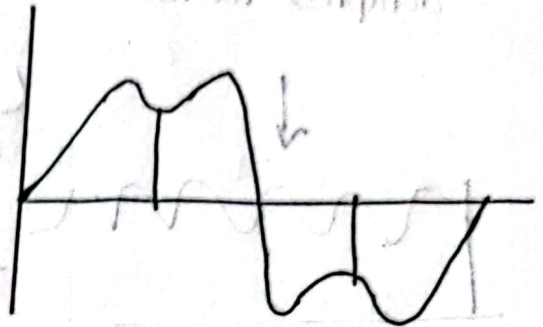


step-1

fundament + 3rd harmonic

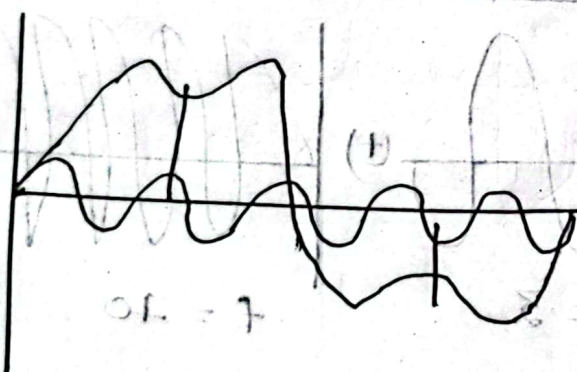


=

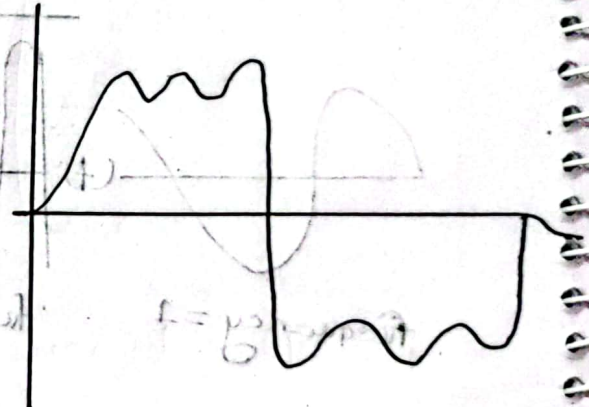


step-2

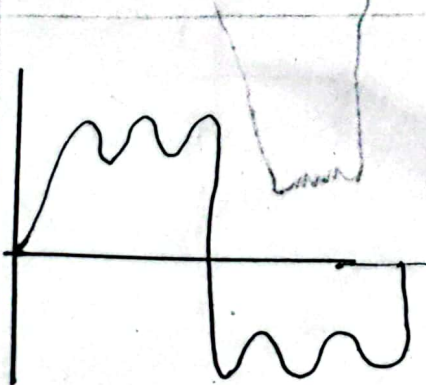
fund.. + 3rd + 5th harmonic



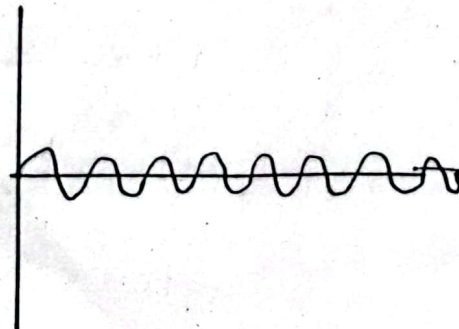
=



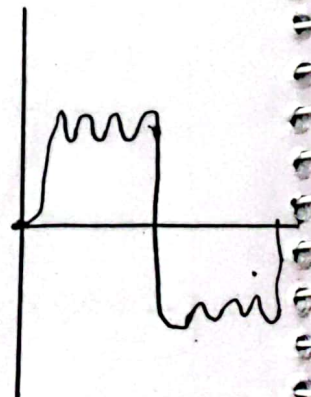
step-3 fund + 3rd + 5th + 7th harmonic



+



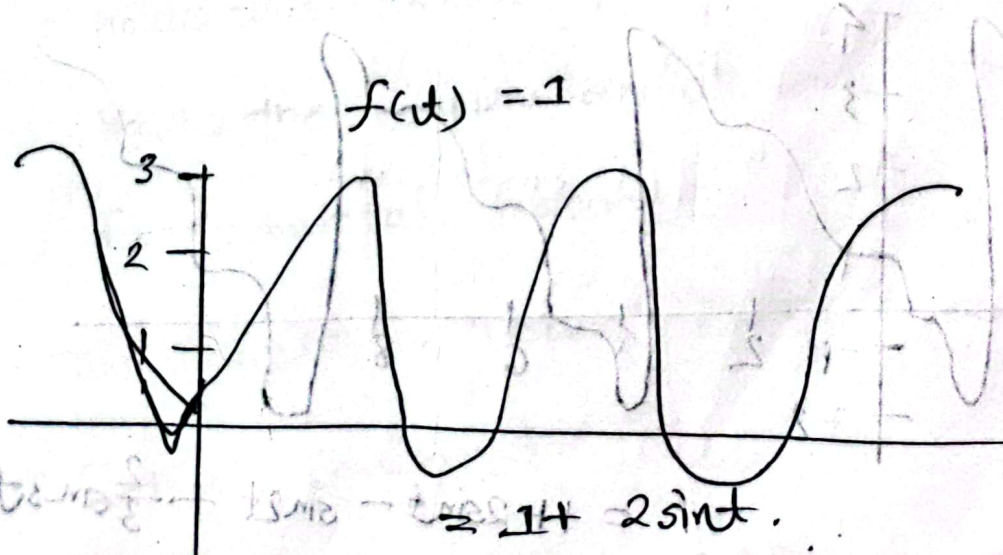
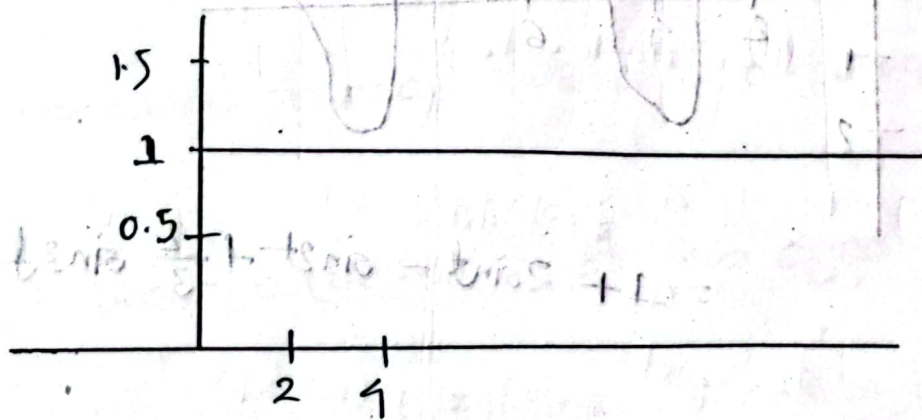
=

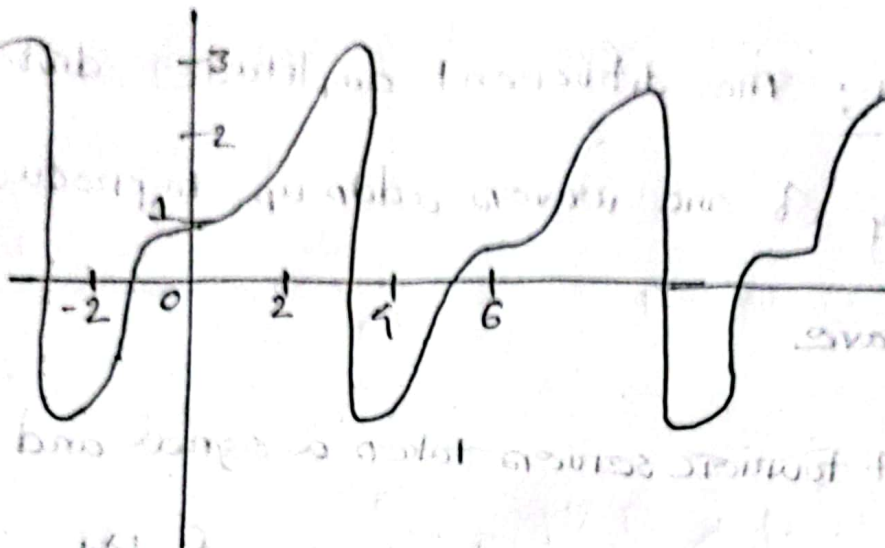


statement: The different amplitude, different frequency of sine waves add up to produce a complex wave

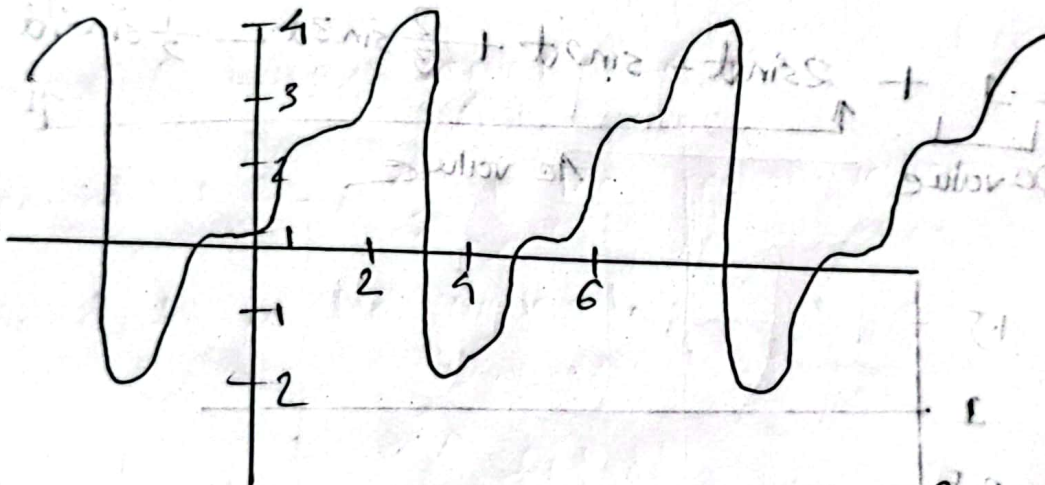
Ex-17 A Fourier series takes a signal and decomposes it into a sum of sines and cosines of different frequencies

$$f(t) = \underbrace{1}_{\text{DC value}} + \underbrace{2\sin t - \sin 2t + \frac{2}{3}\sin 3t - \frac{1}{2}\sin 4t + \frac{2}{5}\sin 5t}_{\text{AC value}} \quad \text{--- T. (i)}$$

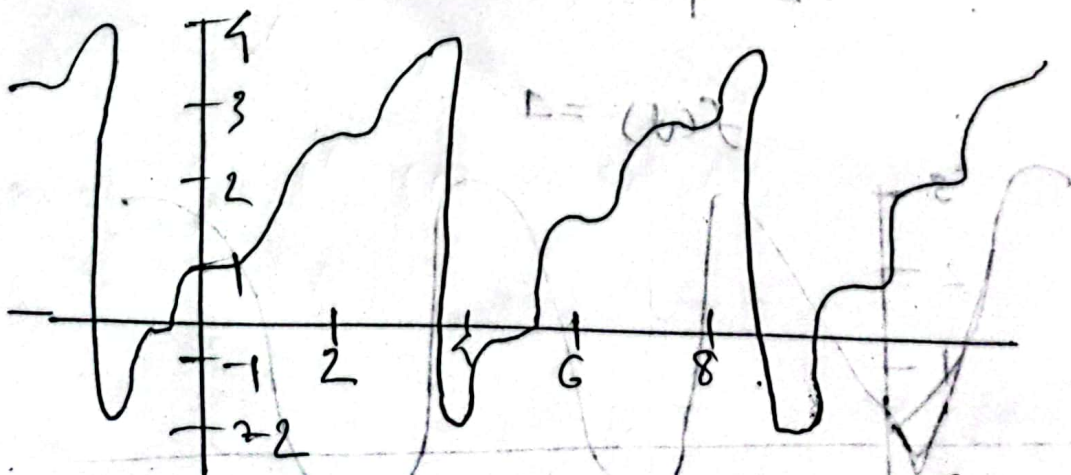




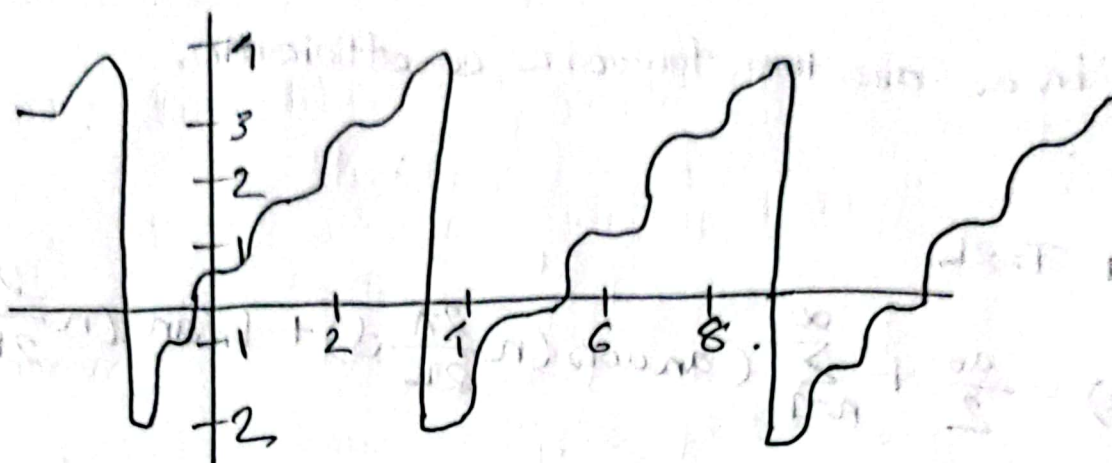
$$f(t) = 1 + 2\sin t - \sin 2t$$



$$= 1 + 2\sin t - \sin 2t + \frac{2}{3}\sin 3t$$



$$f(t) = 1 + 2\sin t - \sin 2t + \frac{2}{3}\sin 3t - \frac{1}{5}\sin 5t$$



$$f(t) = 1 + 2\sin t - \sin 2t + \frac{2}{3}\sin 3t - \frac{1}{2}\sin 4t + \frac{2}{5}\sin 5t$$

$$* f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(n \frac{2\pi}{T} t\right) + b_n \sin\left(n \frac{2\pi}{T} t\right) \right)$$

$$\left[\because \omega = \frac{2\pi}{T} \right]$$

This is called Fourier series of a periodic function $f(t)$.

$\frac{a_0}{2}$ = average (DC) value of the signal

a_n holds the amplitudes of cosine wave

b_n = holds the amplitudes of sine wave.

ω = fundamental frequency

T = This waveform repeats every T seconds where,

T is the period of $f(t)$.

a_n, b_n, a_0 are the fourier co-efficients.

Let $T=2L$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(n \frac{2\pi}{2L} t\right) + b_n \sin\left(n \frac{2\pi}{2L} t\right) \right)$$

$$= \underbrace{\frac{a_0}{2}}_{\text{dc value}} + \sum_{n=1}^{\infty} \underbrace{\left(a_n \cos\left(\frac{n\pi}{L} t\right) + b_n \sin\left(\frac{n\pi}{L} t\right) \right)}_{\text{ac value}}$$

This is called fourier series of a periodic function $f(t)$

For period $T=2L$; $\frac{a_0}{2}$ is the mean value, sometimes

referred to as the dc level.

Ex-19: $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt$

Ex-20: $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt$

Ex-21: $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt$

Ex-22

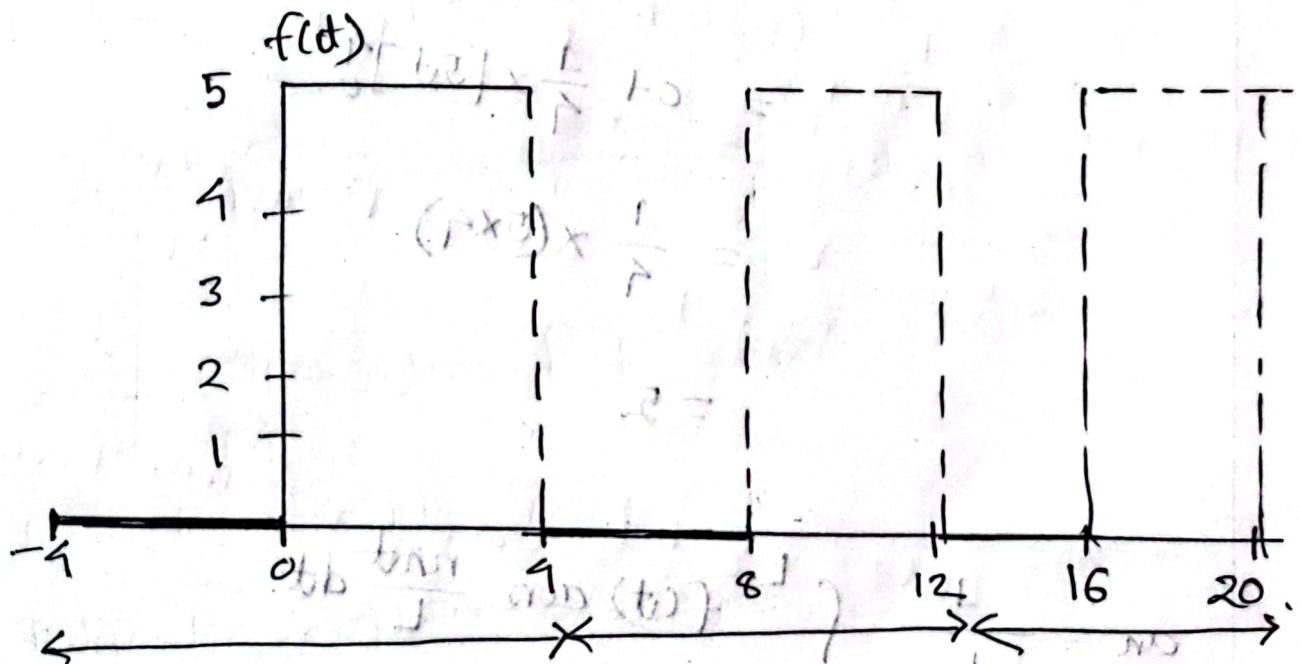
$$y = f(x) = 0, \quad -4 \leq x \leq 0$$

$$y = 5; \quad 0 \leq x \leq 4$$

$$f(x) = f(x+8) \quad \text{Hence, } T = 2L = 8 \quad \therefore L = 4$$

a) sketch the function for 3 cycles

b) find the fourier series for the function.



A periodic signal with period $T = 2L = 8$

We know,

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$

$$= \frac{1}{4} \int_{-4}^4 f(x) dx$$

$$= \frac{1}{4} \int_{-4}^0 f(x) dx + \frac{1}{4} \int_0^4 f(x) dx$$

$$= \frac{1}{4} \int_{-4}^0 0 \cdot dx + \frac{1}{4} \int_0^4 5 dx$$

$$= 0 + \frac{1}{4} \times [5x]_0^4$$

$$= \frac{1}{4} \times (5 \times 4)$$

$$= 5$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx$$

$$= \frac{1}{4} \int_{-4}^4 f(x) \cos \frac{n\pi x}{4} dx$$

$$= \frac{1}{4} \int_{-4}^0 f(x) \cos \frac{n\pi x}{4} dx + \frac{1}{4} \int_0^4 f(x) \cos \frac{n\pi x}{4} dx$$

$$\begin{aligned}
 \therefore a_n &= 0 + \frac{1}{4} \times 5 \int_0^4 \cos \frac{n\pi t}{4} dt \\
 &= \frac{5}{4} \times \frac{4}{n\pi} \left[\sin \frac{n\pi t}{4} \right]_0^4 \\
 &= \frac{5 \times 4}{4n\pi} \left[\sin \frac{4\pi n}{4} - \sin \frac{0 \times n\pi}{4} \right] \\
 &= \frac{5}{n\pi} \sin n\pi
 \end{aligned}$$

$$= 0 \quad \left[\because \sin \pi = \sin 2\pi = \sin 3\pi = \dots = \sin n\pi = 0 \right]$$

Then,

$$\begin{aligned}
 b_n &= \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} dt \\
 &= \frac{1}{4} \int_{-4}^4 f(t) \sin \frac{n\pi t}{4} dt \\
 &= \frac{1}{4} \int_{-4}^0 f(t) \sin \frac{n\pi t}{4} dt + \frac{1}{4} \int_0^4 f(t) \sin \frac{n\pi t}{4} dt \\
 &= 0 + \frac{1}{4} \int_{-4}^0 0 \cdot \sin \frac{n\pi t}{4} dt + \frac{1}{4} \int_0^4 (5) \sin \frac{n\pi t}{4} dt \\
 &= \frac{1}{4} \times \frac{4}{n\pi} \times 5 \times \left(-\cos \frac{n\pi t}{4} \right) \\
 &= \frac{5 \times 4}{4n\pi} \left[\cos \frac{n\pi t}{4} \right]_0^4
 \end{aligned}$$

$$= \frac{-5}{n\pi} \left[\cos \frac{n\pi x}{4} - \cos \frac{n\pi x_0}{4} \right]$$

$$= \frac{-5}{n\pi} [\cos n\pi - 1]$$

The fourier series for the above function:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

$$= \frac{5}{2} + \sum_{n=1}^{\infty} (0) \cos \frac{n\pi x}{L} + \sum_{n=1}^{\infty} -\frac{5}{n\pi} (\cos(n\pi) - 1) \sin \frac{n\pi x}{L}$$

$$= 2.5 - \frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos n\pi - 1) \sin \frac{n\pi x}{4}$$

$$f(x) = 2.5 + \left[\frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos n\pi - 1) \sin \frac{n\pi x}{4} \right]$$

$$\left\{ \frac{1}{1} + \frac{1}{3} + \frac{1}{5} + \dots \right\} \left\{ \frac{1}{1} + \frac{1}{3} + \frac{1}{5} + \dots \right\} = \left\{ \frac{1}{1} + \frac{1}{3} + \frac{1}{5} + \dots \right\}^2$$

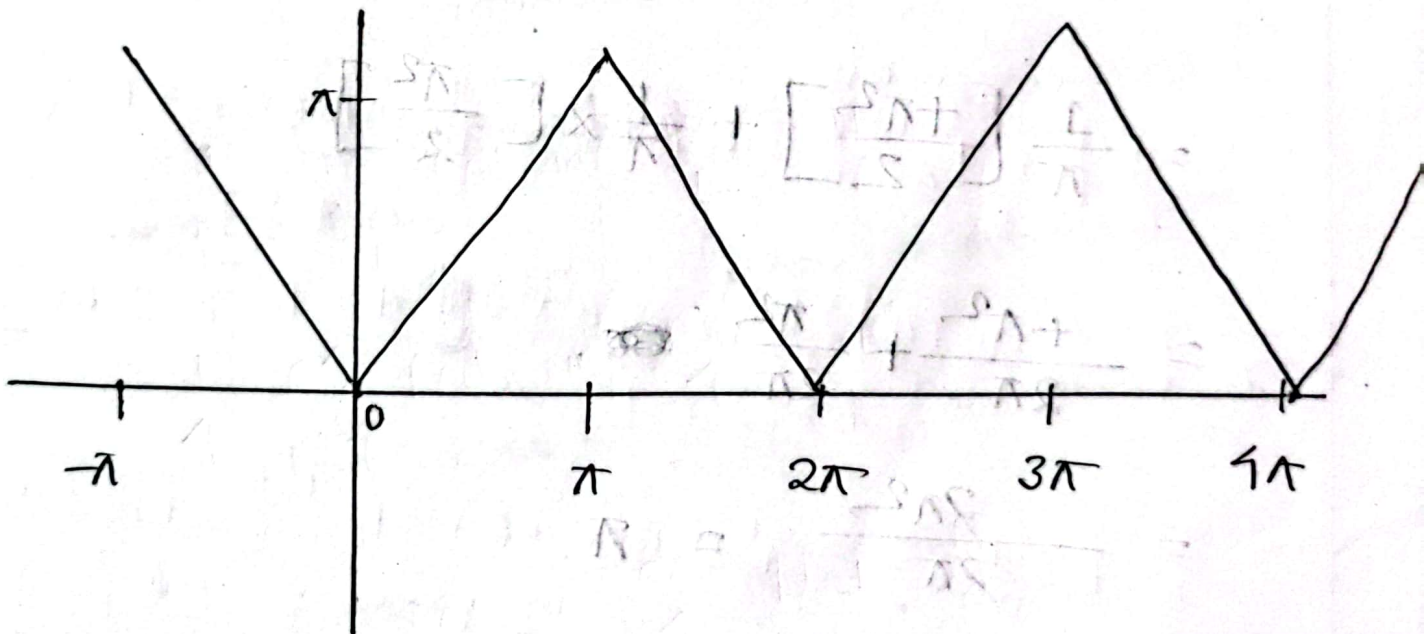
$$y = f(x) = -x, \quad -\pi \leq x \leq 0$$

$$y = x, \quad 0 \leq x \leq \pi$$

$$f(x) = f(x + 2\pi) \quad \text{Here, } T = 2L = 2\pi. \therefore L = \pi.$$

a) sketch the function for 3 cycles.

b) find the fourier series for the function



$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^0 f(x) dx + \frac{1}{\pi} \int_0^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^0 (-x) dx + \frac{1}{\pi} \int_0^{\pi} x dx$$

$$= \frac{1}{\pi} \left[-\frac{x^2}{2} \right]_{-\pi}^0 + \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{0^2}{2} - \frac{-(-\pi)^2}{2} \right] + \frac{1}{\pi} \left[\frac{\pi^2}{2} - 0 \right]$$

$$= \frac{1}{\pi} \left[\frac{+\pi^2}{2} \right] + \frac{1}{\pi} \times \left[\frac{\pi^2}{2} \right]$$

$$= \frac{+\pi^2}{2\pi} + \frac{\pi^2}{2\pi}$$

$$= \frac{2\pi^2}{2\pi} = \pi$$

$$a_0 = \pi$$

$$b_n = 0$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos\left(\frac{n\pi x}{\pi}\right) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^0 f(x) \cos(nx) dx + \frac{1}{\pi} \int_0^{\pi} f(x) \cos(nx) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^0 (-x) \cos(nx) dx + \frac{1}{\pi} \int_0^{\pi} x \cos(nx) dx$$

$$= \frac{-1}{\pi} \int_{-\pi}^0 x \cos(nx) dx + \frac{1}{\pi} \int_0^{\pi} x \cos(nx) dx$$

Now,

$$\text{Let, } I = \int x \cos nx dx \quad \text{--- (ii)}$$

$$= x \sin(nx) - \int \left[\frac{d}{dx}(x) \int \cos nx dx \right] dx$$

$$= x \frac{\sin nx}{n} - \int \left(1 \cdot \frac{\sin nx}{n} \right) dx$$

$$= \frac{x}{n} \sin nx - \left(\frac{1}{n} \cdot \frac{-\cos nx}{n} \right)$$

$$= \frac{x}{n} \sin nx + \frac{1}{n^2} \cos nx \quad \text{--- (iii)}$$

putting the value of (iii) in (ii) $\Rightarrow$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt = \frac{-1}{\pi} \int_{-\pi}^0 t \cos(nt) dt$$

$$+ \frac{1}{\pi} \int_0^{\pi} t \cos(nt) dt$$

$$= -\frac{1}{\pi} \left[\frac{t}{n} \sin(nt) + \frac{1}{n^2} \cos(nt) \right]_{-\pi}^0 + \frac{1}{\pi} \left[\frac{t}{n} \sin(nt) \right.$$

$$\left. + \frac{1}{n^2} \cos(nt) \right]_{\pi}^{\pi}$$

$$= -\frac{1}{\pi} \left[0 + \frac{1}{n^2} \cos(0) - \frac{\pi}{n} \sin(-n\pi) - \frac{1}{n^2} \cos(-n\pi) \right] +$$

$$\frac{1}{\pi} \left[\frac{\pi}{n} \sin(n\pi) + \frac{1}{n^2} \cos(n\pi) \right] + 0 + \frac{1}{n^2} \cos 0$$

$$= \frac{-1}{\pi} \left[\frac{1}{n^2} - 0 - \frac{1}{n^2} \cos(-n\pi) \right] + \frac{1}{\pi} \left[0 + \frac{1}{n^2} \cos n\pi \right]$$

$$= \frac{-1}{\pi n^2} + \frac{1}{\pi n^2} + \frac{1}{\pi n^2} \cos n\pi - \frac{1}{\pi n^2}$$

$$= \frac{2}{\pi n^2} \cos n\pi - \frac{2}{\pi n^2} = \frac{2}{\pi n^2} (\cos n\pi - 1)$$

$$\therefore a_n = \frac{2}{n\pi^2} \cos n\pi - \frac{2}{n\pi^2}.$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx.$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin \frac{n\pi x}{\pi} dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx.$$

$$= \frac{1}{\pi} \int_{-\pi}^0 f(x) \sin nx dx + \frac{1}{\pi} \int_0^{\pi} f(x) \sin nx dx$$

$$= \frac{1}{\pi} \int_{-\pi}^0 -x \sin nx dx + \frac{1}{\pi} \int_0^{\pi} x \sin nx dx$$

$$= -\frac{1}{\pi} \int_{-\pi}^0 x \sin nx dx + \frac{1}{\pi} \int_0^{\pi} x \sin nx dx.$$

— (iv)

Now, $\int x \sin(nx) dx$

$$= x \int \sin(nx) dx - \int \left[\frac{d}{dx}(x) \sin(nx) \right] dx$$

$$= -\frac{x \cos nx}{n} - \int \left[\frac{1(-\cos nx)}{n} \right] dx$$

$$= -\frac{d}{n} \cos(nt) + \int \frac{\sin(nt)}{dz} \quad \text{--- (v)}$$

putting the value of (v) in (iv).

$$b_n = -\frac{1}{\pi} \left[-\frac{d}{n} \cos(nt) + \frac{1}{n^2} \sin(nt) \right]_0^\pi + \frac{1}{\pi}$$

$$\left[\frac{d}{n} \cos(n\pi) + \frac{1}{n^2} \sin(n\pi) \right]_0^\pi$$

$$= -\frac{1}{\pi} \left[0 + \frac{1}{n^2} \sin 0 - \frac{(-d)}{n} \cos(-n\pi) + \right]$$

$$\frac{1}{n^2} \sin(-n\pi) \left] + \frac{1}{\pi} \left[-\frac{d}{n} \cos(n\pi) + \frac{1}{n^2} \sin(n\pi) \right]_0^\pi$$

$$= \frac{d}{\pi} \cos(n\pi) + \frac{-d}{\pi n} \cos(n\pi)$$

(vi)

$$= 0$$

$$b_n = 0$$

$$\left[\frac{1}{n} \cos(n\pi) - \frac{1}{n} \cos(0) \right]$$

The fourier series for the above function is:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}x\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right)$$

$$= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi} (\cos n\pi - 1) \cos\left(\frac{n\pi}{L}x\right) + 0$$

$$= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi} (\cos n\pi - 1) \cos\left(\frac{n\pi}{A}x\right)$$

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi} (\cos n\pi - 1) \cos nx$$

complex way

Ex-39

A plot showing each of the harmonic amplitudes in the wave is called the line spectrum. Plot the line spectrum for the fourier series:

$$f(t) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{1}{2n-1} \cos n\omega t + \sum_{n=1}^{\infty} \frac{(-1)^n}{2n} \sin n\omega t$$

Soln:

we have the fourier series $f(t)$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(n\omega t) + b_n \sin(n\omega t))$$

We can see from the series that,

$$a_n = \frac{1}{2n-1} \quad b_n = \frac{(-1)^n}{2n}$$

$$\text{Now, } R_n = \sqrt{a_n^2 + b_n^2}$$

$$\text{Let, } R_n = a_n \sqrt{a_n^2 + b_n^2}$$

$a_n = \frac{1}{2n-1}$	$b_n = \frac{(-1)^n}{2n}$	$c_n = \sqrt{a_n^2 + b_n^2}$
$a_1 = 1$	$b_1 = -\frac{1}{2}$	$c_1 = \sqrt{1^2 + (-\frac{1}{2})^2} = 1.118$
$a_2 = \frac{1}{3}$	$b_2 = \frac{1}{4}$	$c_2 = \sqrt{(\frac{1}{3})^2 + (\frac{1}{4})^2} = 0.414$
$a_3 = \frac{1}{5}$	$b_3 = -\frac{1}{6}$	$c_3 = \sqrt{(\frac{1}{5})^2 + (-\frac{1}{6})^2} = 0.260$
$a_4 = \frac{1}{7}$	$b_4 = \frac{1}{8}$	$c_4 = \sqrt{(\frac{1}{7})^2 + (\frac{1}{8})^2} = 0.190$

from, here, $n\omega = n$

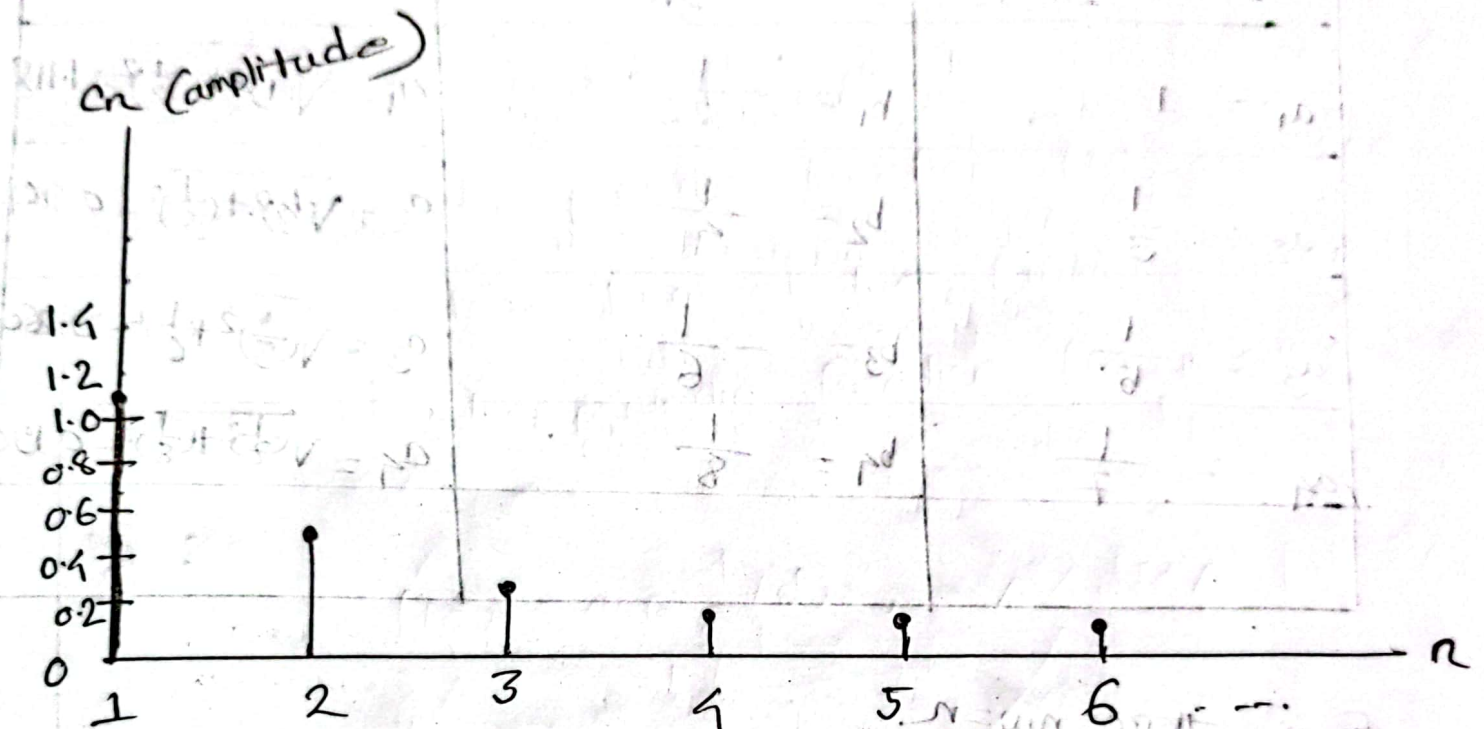
for $n=1$, fundamental frequency = 1st harmonic
 $= \omega = 1$

$n=2$; 2nd harmonic = $2\omega = 2$.

$n=3$ 3rd " = $3\omega = 3$

$n=4$, 4th " = $4\omega = 4$.

The resulting line spectrum:



frequency domain

Compact Trigonometric Fourier series

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\omega t) + \sum_{n=1}^{\infty} b_n \sin(n\omega t) \quad \text{--- (1)}$$

The trigonometric Fourier series equation no (1) contains sine and cosine terms of the same frequency. We can combine the two frequency in a single term trigonometric identity;

$$a_n \cos(n\omega t) + b_n \sin(n\omega t) = c_n \cos(n\omega t + \theta_n) \quad \text{--- (2)}$$

$$\text{where, } c_n = \sqrt{a_n^2 + b_n^2} \quad \text{--- (3)}$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega t}$$

$c_0 \Rightarrow$

where,

$$c_0 = \frac{1}{T} \int_{-L}^L f(t) dt$$

$$c_n = \frac{1}{T} \int_{-L}^L f(t) e^{-in\omega t} dt$$

$$c_n^* = c_{-n} = \frac{1}{T} \int_{-L}^L f(t) e^{in\omega t} dt$$

Ex-37: find the complex form of the fourier series expansion of the periodic function $f(t)$ and find the trigonometric form of the fourier series expansion of the periodic function $f(t)$ is given

by :

$$f(t) = \cos \frac{t}{2} ; -\pi < t < \pi \quad [T = 2\pi]$$

Soln:

The complex form of the fourier series expansion of a periodic function $f(t)$ of period T ,

is :

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega t} \quad \text{--- (i)}$$

$$\left. \begin{aligned} \text{for } T &= 2\pi \\ \omega &= \frac{2\pi}{T} = 1 \end{aligned} \right\}$$

where,

$$c_0 = \frac{1}{T} \int_{-L}^L f(t) dt.$$

$$c_n = \frac{1}{T} \int_{-L}^L f(t) e^{-in\omega t} dt.$$

$$c_n^* = c_{-n} = \frac{1}{T} \int_{-L}^L f(t) e^{in\omega t} dt.$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-in\omega t} dt.$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos \frac{1}{2} t e^{-in \frac{2\pi}{T} t} dt$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos \frac{1}{2} t e^{-in \frac{2\pi}{2\pi} t} dt$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos \frac{1}{2} t e^{-int} dt$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left\{ \frac{1}{2} (e^{i(\frac{1}{2}t)} + e^{-i(\frac{1}{2}t)}) \right\} e^{-int} dt$$

$$\therefore \cos u = \frac{1}{2} (e^{iu} + e^{-iu})$$

$$= \frac{1}{4\pi} \int_{-\pi}^{\pi} \{ e^{i(\frac{1}{2}t)} \cdot e^{-int} + e^{-i(\frac{1}{2}t)} \cdot e^{int} \} dt$$

$$= \frac{1}{4\pi} \int_{-\pi}^{\pi} \{ e^{i(\frac{1}{2}t - nt)} + e^{-i(\frac{1}{2}t + nt)} \} dt$$

$$= \frac{1}{4\pi} \int_{-\pi}^{\pi} \{ e^{i(\frac{t-2nt}{2})} + e^{-i(\frac{t+2nt}{2})} \} dt$$

$$= \frac{1}{4\pi} \int_{-\pi}^{\pi} \{ e^{i(\frac{1-2n}{2})t} + e^{-i(\frac{1+2n}{2})t} \} dt$$



$$= \frac{1}{4\pi} \int_{-\pi}^{\pi} \left\{ e^{-i(\frac{2n-1}{2})t} + e^{-i(\frac{2n+1}{2})t} \right\} dt$$

$$= \frac{1}{4\pi} \left[\frac{e^{-i(\frac{2n-1}{2})t}}{-\left(\frac{2n-1}{2}\right)} + \frac{e^{-i(\frac{2n+1}{2})t}}{-\left(\frac{2n+1}{2}\right)} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{4\pi} \left[\frac{2e^{-i\left(\frac{2n-1}{2}\right)t}}{-i(2n-1)} + \frac{2e^{-i\left(\frac{2n+1}{2}\right)t}}{-i(2n+1)} \right]_{-\pi}^{\pi}$$

$$= \frac{-2}{4\pi} \left[\frac{e^{-int} \cdot e^{\frac{it}{2}}}{i(2n-1)} + \frac{e^{-int} \cdot e^{-\frac{it}{2}}}{i(2n+1)} \right]_{-\pi}^{\pi}$$

$$= \frac{-2}{4\pi} \left[\frac{e^{-in\pi} \cdot e^{\frac{i\pi}{2}}}{i(2n-1)} + \frac{e^{-in\pi} \cdot e^{-\frac{i\pi}{2}}}{i(2n+1)} - \frac{e^{in\pi} \cdot e^{-\frac{i\pi}{2}}}{i(2n-1)} - \frac{e^{in\pi} \cdot e^{\frac{i\pi}{2}}}{i(2n+1)} \right]$$

(ii)

we have,

$$e^{ix} = \cos x + i \sin x$$

$$\therefore e^{\frac{i\pi}{2}} = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$$

$$= 0 + i \cdot 1 = i$$

And,

$$= e^{-i\frac{\pi}{2}} \left(1 \cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right)$$

$$= 0 - i \cdot 1 = -i$$

$$e^{in\pi} = \cos n\pi + i \sin n\pi$$

$$= (-1)^n + 0 = (-1)^n$$

Since, if $n=1$ $\cos n\pi = -1$

$$n=2 \quad \cos n\pi = 1$$

$$n=3 \quad \cos n\pi = -1$$

Hence we can write

$$\cos n\pi = (-1)^n = n[1, 2, 3, \dots]$$

Again $e^{in\pi} = \cos n\pi - i \sin n\pi$

$$= (-1)^n - i \cdot 0$$

$$= (-1)^n$$

putting the value into eq - ii,

$$a_n = \frac{-2}{4\pi} \left[\frac{e^{-in\pi} e^{\frac{i\pi}{2}}}{i(2n-1)} + \frac{e^{in\pi} e^{-\frac{i\pi}{2}}}{i(2n+1)} - \frac{e^{in\pi} e^{\frac{i\pi}{2}}}{i(2n-1)} - \frac{e^{-in\pi} e^{-\frac{i\pi}{2}}}{i(2n+1)} \right]$$

~~$$= \frac{-2}{4\pi} \left[\frac{e^{-in\pi} e^{\frac{i\pi}{2}}}{i(2n-1)} + \frac{e^{in\pi} e^{-\frac{i\pi}{2}}}{i(2n+1)} - \frac{e^{in\pi} e^{\frac{i\pi}{2}}}{i(2n-1)} - \frac{e^{-in\pi} e^{-\frac{i\pi}{2}}}{i(2n+1)} \right]$$~~

$$= \frac{-2}{4\pi} \left[\frac{e^{-in\pi} \cdot i}{i(2n-1)} + \frac{e^{-in\pi} (-i)}{i(2n+1)} - \frac{e^{in\pi} (-i)}{i(2n-1)} - \frac{e^{in\pi} (i)}{i(2n+1)} \right]$$

$$= \frac{-2}{4\pi} \left[\frac{(-1)^n \cdot i}{i(2n-1)} + \frac{(-1)^n (-i)}{i(2n+1)} - \frac{(-1)^n (-i)}{i(2n-1)} - \frac{(-1)^n (i)}{i(2n+1)} \right]$$

$$= \frac{-2}{4\pi} \left[\frac{(-1)^n}{(2n-1)} - \frac{(-1)^n}{(2n+1)} + \frac{(-1)^n}{(2n-1)} - \frac{(-1)^n}{(2n+1)} \right]$$

$$= \frac{-2 (-1)^n}{4\pi} \left[\frac{1}{(2n-1)} - \frac{1}{(2n+1)} + \frac{1}{(2n-1)} - \frac{1}{(2n+1)} \right]$$

$$= \frac{-2 \cdot (-1)^n}{4\pi} \left[\frac{2}{(2n-1)} - \frac{2}{(2n+1)} \right]$$

$$= \frac{-2 \cdot (-1)^n}{4\pi} \left[\frac{4n+2 - 4n+2}{(2n-1)(2n+1)} \right]$$

$$= \frac{(-2)(-1)^n}{\cancel{\pi}} \times \left[\frac{1}{4n^2-1} \right]$$

$$= \frac{2(-1)^{n+1}}{\pi} \times \frac{1}{(4n^2-1)}$$

$$\therefore c_n = \frac{2(-1)^{n+1}}{\pi} \times \frac{1}{(4n^2-1)}$$

From (i), the complex form of the fourier series expansion of a periodic function $f(t)$, of period T , is:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega t}$$

$$= \sum_{n=-\infty}^{\infty} \left\{ \frac{2}{\pi} \left[\frac{1}{4n^2-1} \times (-1)^{n+1} \right] e^{in\omega t} \right\}$$

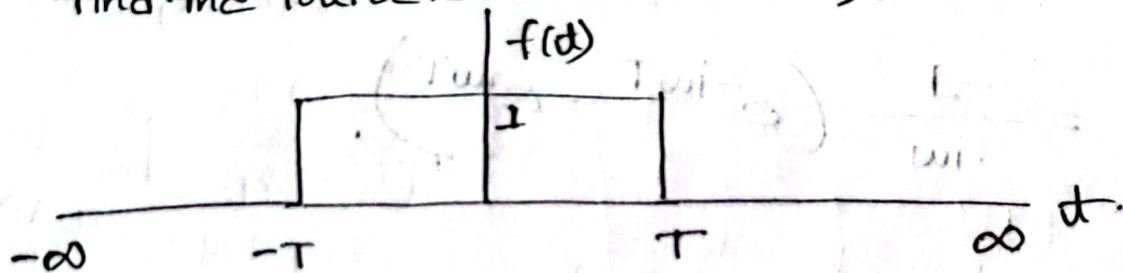
$$= \sum_{n=-\infty}^{\infty} \left\{ \frac{2}{\pi} \left[\frac{1}{4n^2-1} \right] (-1)^{n+1} \right\} e^{in\omega t}$$

(A)

Example : 39 Consider the rectangular pulse figure

$$f(t) = \begin{cases} 1 & -T \leq t \leq T \\ 0 & |t| > T \end{cases}$$

find the fourier transform $f(t)$.



we have,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} [g(\omega)] e^{i\omega t} d\omega.$$

where,

$$g(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt.$$

$$= \int_{-\infty}^{-T} f(t) e^{-i\omega t} dt + \int_{-T}^T f(t) e^{-i\omega t} dt + \int_T^{\infty} f(t) e^{-i\omega t} dt$$

$$= \int_{-\infty}^{-T} 0 \cdot e^{-i\omega t} dt + \int_{-T}^T 1 \cdot e^{-i\omega t} dt + \int_T^{\infty} 0 \cdot e^{-i\omega t} dt$$

$$= \int_{-T}^T e^{-i\omega t} dt.$$

$$= \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{-T}^T.$$

$$= \frac{-1}{i\omega} (e^{-i\omega T} - e^{i\omega T}).$$

$$= \frac{1}{i\omega} (e^{i\omega T} - e^{-i\omega T}).$$

$$= \frac{2}{\omega} \left\{ \frac{1}{2i} (e^{i\omega T} - e^{-i\omega T}) \right\}$$

$$= \frac{2}{\omega} \sin(\omega T) \quad \text{--- (i)}$$

$$\left[\because \sin x = \frac{1}{2i} (e^{ix} - e^{-ix}) \right]$$

$$\text{OR, [since we have } \sin x = \frac{1}{2i} (e^{ix} - e^{-ix})]$$

$$\therefore g(\omega) = \frac{2}{\omega} \sin(\omega T).$$

$$= \frac{2T}{\omega T} \sin(\omega T)$$

$$g(\omega) = 2T \frac{\sin(\omega T)}{\omega T}$$

Then we write for all w , $g(w) = 2T \frac{\sin(wT)}{wT}$ (ii)

for $w=0$, the integral simplifies to $2T$. It is straight forward to show using L'Hospital's rule that, from (i).

$$\lim_{w \rightarrow 0} \frac{2}{w} \sin(wT)$$

$$= \lim_{w \rightarrow 0} \frac{2 \sin(wT)}{w}$$

$$= \lim_{w \rightarrow 0} \frac{2 \sin(wT) \cdot T}{1}$$

$$= \frac{2 \cos(0) \cdot T}{1}$$

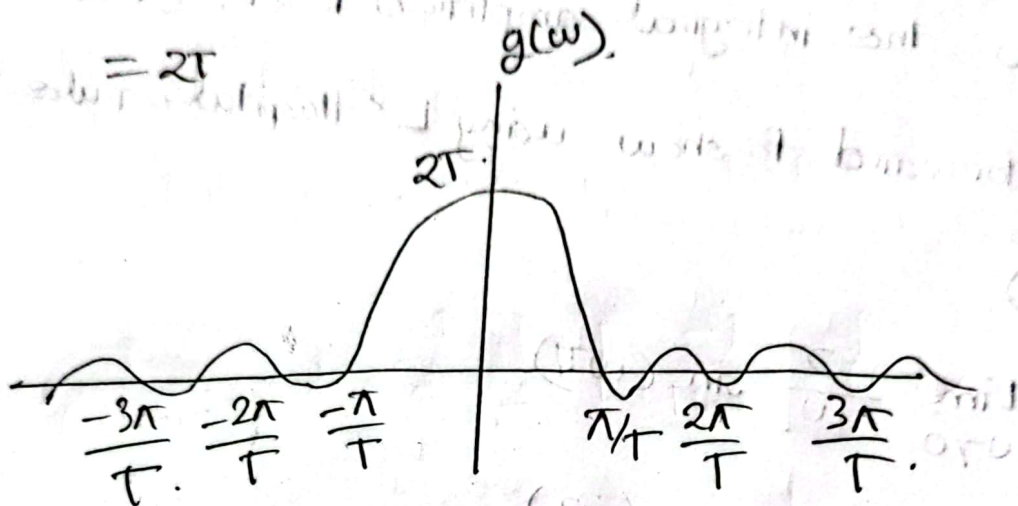
$$= 2T.$$

from (ii),

$$g(w) = 2T \frac{\sin(wT)}{wT}$$

$$\lim_{w \rightarrow 0} g(w) = \lim_{w \rightarrow 0} 2T \frac{\sin(wT)}{wT}$$

$$= 2T \lim_{\omega \rightarrow 0} \frac{\sin(\omega T)}{\omega T} \quad \left[\because \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \right]$$



fourier transform of $f(t)$.

Ex-42: fourier transform of $f(t) = e^{-|t|}$.

OR find fourier transform of

$$f(t) = e^{-t} \quad ; t > 0.$$

$$= e^t \quad ; t < 0.$$

Ans: Given

$$f(t) = e^{-t} \quad ; t > 0$$

$$= e^t \quad ; t < 0$$

We have,

$$g(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$= \int_{-\infty}^0 f(t) e^{-i\omega t} dt + \int_0^{\infty} f(t) e^{-i\omega t} dt$$

$$= \int_{-\infty}^0 e^t e^{-i\omega t} dt + \int_0^{\infty} e^{-t} e^{-i\omega t} dt$$

$$= \int_{-\infty}^0 e^{t-i\omega t} dt + \int_0^{\infty} e^{-(t+i\omega t)} dt$$

$$= \int_{-\infty}^0 e^{(1-i\omega)t} dt + \int_0^{\infty} e^{-(1+i\omega)t} dt$$

$$= \left[\frac{e^{(1-i\omega)t}}{(1-i\omega)} \right]_{-\infty}^0 + \left[\frac{e^{-(1+i\omega)t}}{-(1+i\omega)} \right]_0^{\infty}$$

$$= \left[\frac{e^{(1-i\omega) \cdot 0}}{(1-i\omega)} - \frac{e^{(1-i\omega)(-\infty)}}{(1-i\omega)} \right] + \left[\frac{e^{-(1+i\omega)\infty}}{-(1+i\omega)} - \frac{e^{-(1+i\omega) \cdot 0}}{-(1+i\omega)} \right]$$

$$= \left[\frac{e^0}{(1-i\omega)} - \frac{e^{-\infty}}{(1-i\omega)} \right] + \left[\frac{e^{-\infty}}{(1+i\omega)} - \frac{e^0}{(1+i\omega)} \right]$$

$$h_b = \left[\frac{1}{1-i\omega} - \frac{1}{\infty(1-i\omega)} \right] + \frac{-1}{(1+i\omega)} \left[\frac{1}{\infty} - \frac{1}{e^0} \right]$$

$$h_b = \frac{1}{(1-i\omega)} \left[1 - \frac{1}{\infty} \right] + \left[\frac{-1}{(1+i\omega)} \left\{ \frac{1}{\infty} - \frac{1}{1} \right\} \right]$$

$$= \frac{1}{(1-i\omega)} (1-0) + \frac{-1}{(1+i\omega)} (0-1)$$

$$= \frac{1}{(1-i\omega)} + \frac{1}{(1+i\omega)}$$

$$\frac{1 + 1 - i\omega + i\omega}{1 - i^2\omega^2}$$

$$= \frac{2}{1 + \omega^2}$$

$$\left[\frac{g(\omega)^2}{1 + \omega^2} \right] \cdot \frac{A}{(1-i)}$$

Ex-55

find the laplace transform of $f(t)=1$.

we have,

$$L(f(t)) = \int_0^{\infty} f(t) e^{-st} dt.$$

given, $f(t)=1$

$$\therefore L(1) = \int_0^{\infty} 1 \cdot e^{-st} dt$$

$$= \int_0^{\infty} \left[\frac{e^{-st}}{-s} \right]_0^{\infty}$$

$$= -\frac{1}{s} [e^{-\infty} - e^{-0}]$$

$$= -\frac{1}{s} \left[\frac{1}{e^{\infty}} - \frac{1}{e^0} \right]$$

$$= \frac{-1}{s} \left[-\frac{1}{\infty} - \frac{1}{1} \right]$$

$$= \frac{-1}{s} [0 - 1]$$

$$= \frac{1}{s}$$

$$\therefore L(f(t)) = L(1) = \frac{1}{s}$$

$$L(f(t)) = L(e^{at}) = \frac{1}{s-a}$$

Ex-58

$$L(t^n) = \frac{n!}{s^{n+1}}$$

$$\Rightarrow L(\sin at) = \frac{a}{s^2 + a^2}$$

$$L(\cos at) = \frac{s}{s^2 + a^2}$$

$$L(t) = \frac{1}{s^2}$$

The first Shift Theorem

We have seen that a Laplace transform of $f(t)$

is a function of s only, i.e.

$$L\{f(t)\} = f(s)$$

The first step theorem states that

$$\text{If } L\{f(t)\} = f(s). \quad \text{--- (i)}$$

$$\text{Then } L\{e^{-at} f(t)\} = f(sta) \quad \text{--- (ii)}$$

Ex-64

$$\text{find } L\{e^{-at} t\} = ?$$

$$\text{Hence, } f(t) = t$$

The first shift theorem states that,

$$\text{If } L\{f(t)\} = f(s). \quad \text{--- (i)}$$

$$\text{Then } L\{e^{-at} f(t)\} = f(sta) \quad \text{--- (ii)}$$

we have, according to the equation (i),

$$L\{f(t)\} = L(t) = \frac{1}{s^2}$$

$$\text{If } f(s) = \frac{1}{s^2}$$

$$\therefore f(sta) = \frac{1}{(sta)^2}$$

Hence according to eqn (ii) we can write,

$$L\{e^{-at} f(t)\} = f(sta) = \frac{1}{(sta)^2} \quad \text{--- (A)}$$

Ex-69 find $L\{e^{-4t} d^2\} = ?$

we have,

$$f(t) = t^2$$

According to first shift theorem,

$$\text{If } L\{f(t)\} = f(s) \text{ --- (i)}$$

$$\text{The } L\{e^{-4t} f(t)\} = f(s+4) \text{ --- (ii)}$$

from eqn - (i) we can write

$$L(f(t)) = L(t^2) = \frac{2!}{s^{2+1}} = \frac{2!}{s^3} = f(s)$$

$$\therefore f(s+4) = \frac{2!}{(s+4)^3}$$

Now, from eqn - (ii) we can write

$$L\{e^{-4t} t^2\} = f(s+4) = \frac{2!}{(s+4)^3} = \frac{2!}{(s+4)^3}$$

Ex-76

Prove that,

$$L(f''(t)) = s^2 L\{f(t)\} - sf(0) - f'(0).$$

we have,

$$L(f(t)) = \int_0^{\infty} f(t) e^{-st} dt \quad \text{--- (i)}$$

$$\therefore L(f'(t)) = \int_0^{\infty} f'(t) e^{-st} dt.$$

$$\left[f(t) = f'(t) \right] \rightarrow \text{--- (ii)}$$

$$\text{Now, } \int f'(t) e^{-st} dt.$$

$$= e^{-st} \int f'(t) dt = \int \left\{ \frac{d}{dt} (e^{-st}) \int f'(t) dt \right\} dt$$

$$= e^{-st} f(t) - \int \{ s e^{-st} f(t) \} dt$$

$$= e^{-st} f(t) + \int \{ s (e^{-st}) f(t) \} dt$$

$$= e^{-st} f(t) + s \int \{ (e^{-st}) f(t) \} dt.$$

Now,

$$= [e^{-st} f(t)]_0^{\infty} + s \int_0^{\infty} e^{-st} f(t) dt$$

$$= [e^{-s \times \infty} f(\infty) - e^{-s \times 0} f(0)] + s \int_0^{\infty} e^{-st} f(t) dt$$

$$= [e^{-\infty} f(\infty) - e^0 f(0)] + s \int_0^{\infty} e^{-st} f(t) dt$$

$$= [\frac{1}{\infty} f(\infty) - 1 f(0)] + s \int_0^{\infty} e^{-st} f(t) dt$$

$$= [0 \cdot f(\infty) - f(0)] + s \int_0^{\infty} e^{-st} f(t) dt$$

$$= -f(0) + s \int_0^{\infty} e^{-st} f(t) dt$$

$$= -f(0) + s L \{f(t)\}$$

$$\therefore L \{f'(t)\} = s L \{f(t)\} - f(0)$$

————— (iii)

Now,

replacing $f(t)$ by $f'(t)$ and $f'(t)$ by $f''(t)$ in (iii)

we get,

$$L(f'(t)) = sL\{f(t)\} - f(0).$$

$$L(f''(t)) = sL\{f'(t)\} - f'(0). \quad \text{--- (iv)}$$

putting the value $L(f'(t))$ from (iii) in (iv) we get

$$L(f''(t)) = sL\{f'(t)\} - f'(0).$$

$$= s[sL\{f(t)\} - f(0)] - f'(0).$$

$$= s^2L\{f(t)\} - sf(0) - f'(0).$$

$$\therefore L(f''(t)) = s^2L\{f(t)\} - sf(0) - f'(0) \quad \text{(proved)}$$

Similarly

$$\therefore L(f'''(t)) = s^3L\{f(t)\} - s^2f(0) - sf'(0) - f''(0).$$

$$\therefore L(f^{(4)}(t)) = s^4L\{f(t)\} - s^3f(0) - s^2f'(0) - sf''(0) - f'''(0).$$

Ex-22 Express the following function in terms of unit step function and find its Laplace transform.

$$f(t) = \begin{cases} 8; & t < 2 \\ 6; & t > 2 \end{cases}$$

Soln:

we have,

$$L(u(t-a)) = \frac{e^{-as}}{s}$$

$$\therefore L(u(t-2)) = \frac{e^{-2s}}{s} = \frac{1}{s} - \frac{1}{s} + \frac{1}{s} \quad \text{--- (1)}$$

Given,

$$f(t) = \begin{cases} 8; & t < 2 \\ 6; & t > 2 \end{cases}$$

$$f(t) = \begin{cases} 8+0; & t < 2 \\ 8-2; & t > 2 \end{cases}$$

$$= 8 + \begin{cases} 0; & t < 2 \\ -2; & t > 2 \end{cases}$$

$$= 8 + (-2) \begin{cases} 0; & t < 2 \\ 1; & t > 2 \end{cases}$$

$$= 8 + (-2) \begin{cases} 1; & t > 2 \\ 0; & t < 2 \end{cases}$$

$$f(t) = 8 + (-2)u(t-2).$$

$$f(t) = 8 - 2u(t-2).$$

$$L(f(t)) = L\{8 - 2u(t-2)\}.$$

$$= L\{8\} - 2L\{u(t-2)\}$$

$$= 8L\{1\} - 2L\{u(t-2)\}.$$

$$\therefore L\{f(t)\} = 8 \frac{1}{s} - 2 \times \frac{e^{-2s}}{s}.$$

$$[\because L(1) = \frac{1}{s} \quad (\text{ex-55}) \quad \& \quad L(u(t-2)) = \frac{e^{-2s}}{s} \quad (\text{ex-21})].$$

Q-103

f-5-7

Solve the following Initial Value Problem (IVP) by Laplace Transform $y'' + y = t$, $y(0) = 1$, $y'(0) = 2$.

Soln:

$$\text{Let } y = f(t)$$

$$y' = f'(t)$$

$$y'' = f''(t)$$

Given,

$$y'' + y = t$$

$$\text{That is, } \frac{d^2 y}{dt^2} + y = t$$

Taking the Laplace transform of both sides the diff. equation and using the given conditions,

we have,

$$y'' + y = t$$

~~$$L\{f''(t)\}$$~~

$$L\{y''\} + L\{y\} = L\{t\}$$

we have,

$$L\{f''(t)\} = s^2 L\{f(t)\} - sf(0) - f'(0)$$

$$\therefore s^2 L\{y\} - sf(0) - f'(0) + L\{y\} = L\{d\}.$$

$$\Rightarrow s^2 y - sf(0) - f'(0) + y = \frac{1}{s^2} \quad \left[\text{Let, } L\{y\} = y \right]$$

$$\Rightarrow s^2 y - s \cdot 1 - (-2) + y = \frac{1}{s^2} \quad \left[\begin{array}{l} y(0) = y(0) = 1 \\ y'(0) = y'(0) = -2 \end{array} \right]$$

$$\Rightarrow s^2 y + y - s + 2 = \frac{1}{s^2}.$$

$$\Rightarrow y(s^2 + 1) = s - 2 + \frac{1}{s^2}$$

$$\Rightarrow y = \frac{s-2}{(s^2+1)} + \frac{1}{s^2(s^2+1)}$$

$$\therefore L\{y\} = y = \frac{s}{s^2+1} - \frac{2}{s^2+1} + \frac{1}{(s^2)^2} - \frac{1}{(s^2+1)}$$

$$= \frac{s}{s^2+1} + \frac{1}{s^2} - \frac{3}{(s^2+1)}$$

$$\therefore L\{y\} = y = \frac{s}{s^2+1} + \frac{1}{s^2} - \frac{3}{s^2+1}$$

$$y = L^{-1}\{y\} = L^{-1}\left\{\frac{s}{s^2+1} + \frac{1}{s^2} - \frac{3}{s^2+1}\right\}.$$

$$(1) \quad y = L^{-1}(y) = L^{-1}\left(\frac{s}{s^2+1}\right) + L^{-1}\left(\frac{1}{s^2}\right) - L^{-1}\left(\frac{3}{s^2+1}\right)$$

$$= \cos t + t - 3 \sin t \quad \underline{\underline{A}}$$

Proof:

$$\therefore y = \cos t + t - 3\sin t.$$

$$\therefore y' = -\sin t + 1 - 3\cos t.$$

$$\therefore y'' = -\cos t + 0 + 3\sin t.$$

$$\begin{aligned}\therefore y'' + y &= -\cos t + 3\sin t + \cos t + t - 3\sin t + t \\ &= t.\end{aligned}$$

Again,

$$\therefore y = \cos t + t - 3\sin t$$

$$y(0) = \cos(0) + 0 - 3\sin 0.$$

$$\therefore y(0) = 1$$

Again,

$$y' = -\sin t + 1 - 3\cos t$$

$$y'(0) = -\sin 0 + 1 - 3\cos 0$$

$$= 0 + 1 - 3$$

$$\therefore y'(0) = -2$$

Convolution Sum

Steps

- i) folding
- ii) shifting
- iii) Multiplication
- iv) summation

1st time :

- i) folding
- ii) multiplication.
- iii) summation

2nd time & more:

- i) shifting
- ii) multiplication
- iii) summation

Ex-102 :-

Evaluate the convolution sums of $y[n] = x[n] * h[n]$.

where,

$$x[n] = 1, n=0 \\ = 3, n=1$$

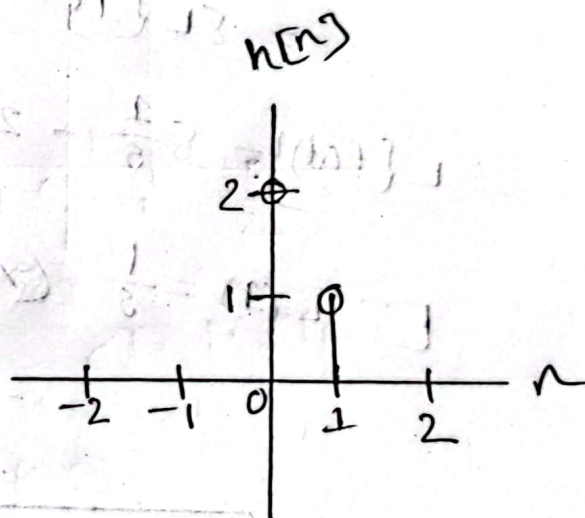
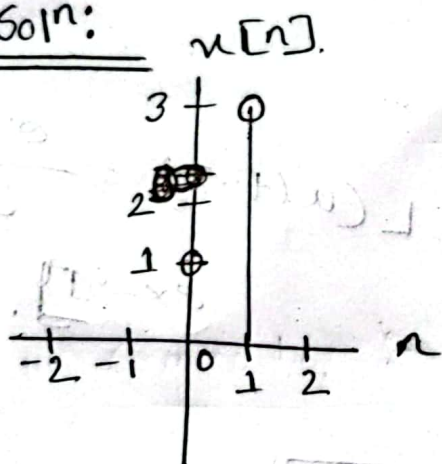
and

$$h[n] = 2, n=0.$$

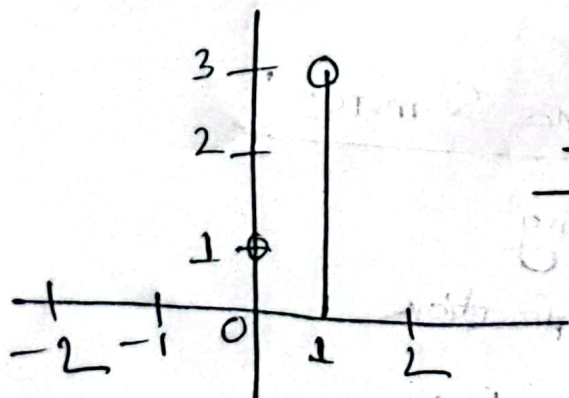
$$= 1, n=1.$$

n represents the time index

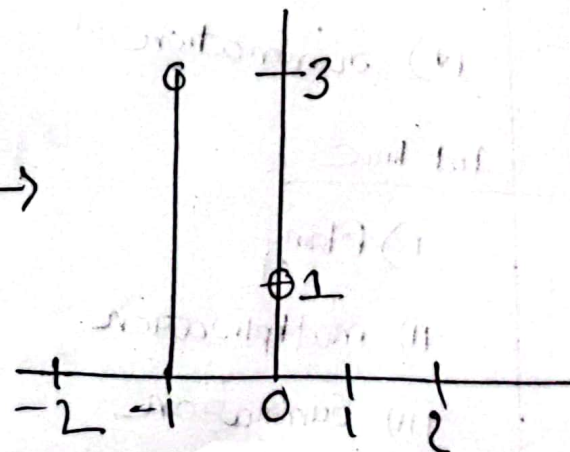
Soln:



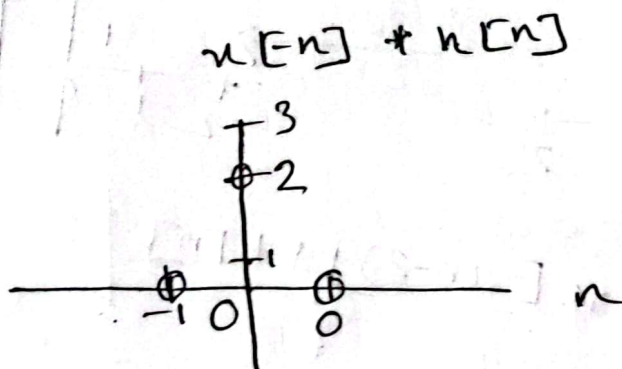
(i) folding: $x[n] = x[-n]$.
i.e, $x[0] = x[0]$.
 $x[1] = x[-1]$.



folding



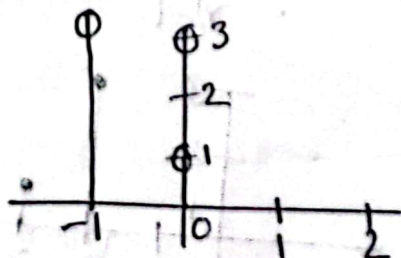
(ii) multiplication



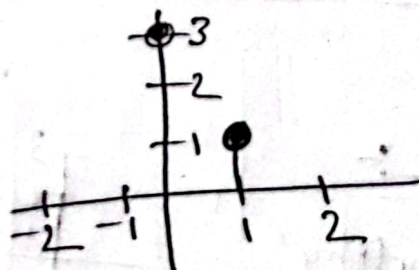
(iii) Summation $y[0] = 0 + 2 + 0 = 2$.

2nd time:

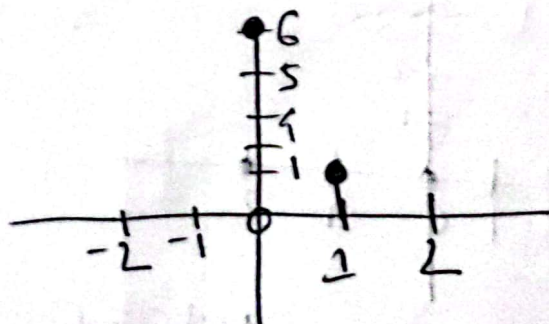
(i) Shifting:



shifting $\rightarrow$



(ii) multiplication:

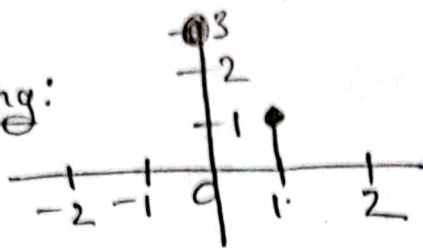


(ii) sum $y[1] = 0 + 6 + 1 + 0 = 7$

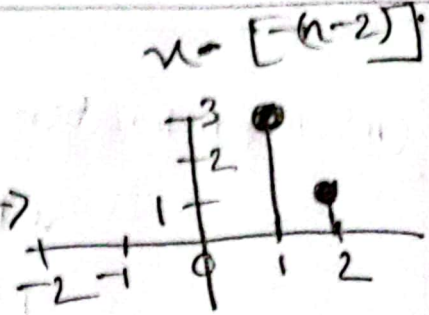
3rd time :-

(i)

Shifting:



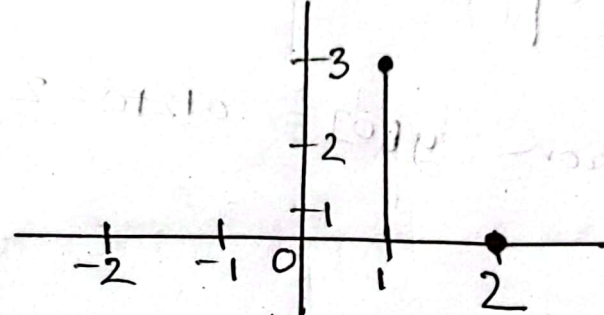
$x[n-1]$



$x[n-2]$

$x[n-2] * h[n]$

(ii) Multiplication:-



(iii)

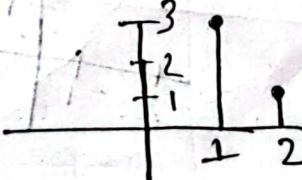
Summation:-

$$y[2] = 0 + 0 + 3 + 0 = 3$$

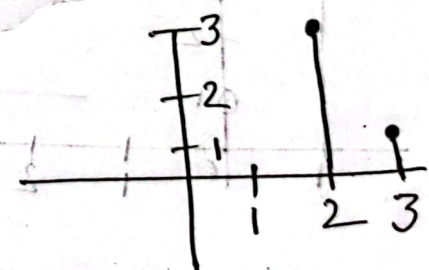
4th time :-

(i)

Shifting:

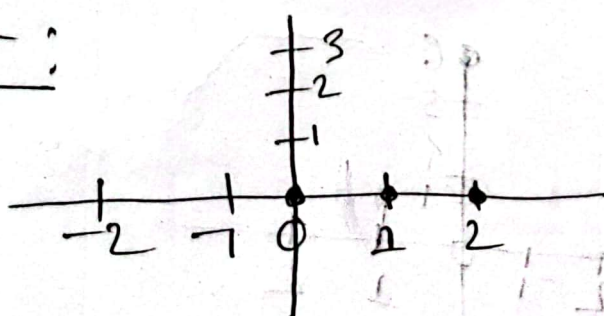


$x[n-2]$



$x[n-3]$

(ii) multiplication :



(iii)

Summation:

$$y[3] = 0 + 0 + 0 + 0 = 0$$

finally we get

$$y[0] = 2, y[1] = 7, y[2] = 3.$$

$$\therefore \text{Convolution Sum } y[n] = [2 \ 7 \ 3].$$

Convolution Integral

- i. folding
- ii. shifting
- iii. finding Area

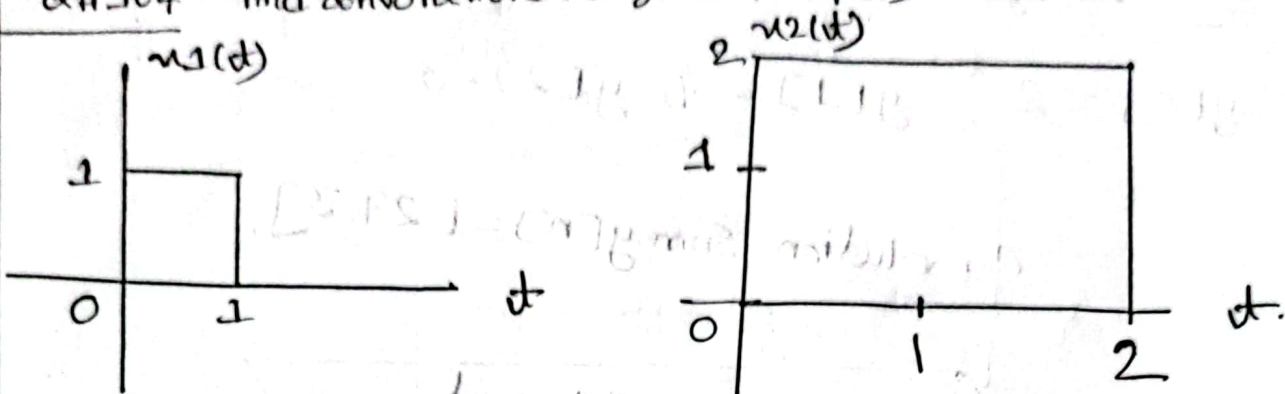
1st time:

- i) folding
- ii) finding Area.

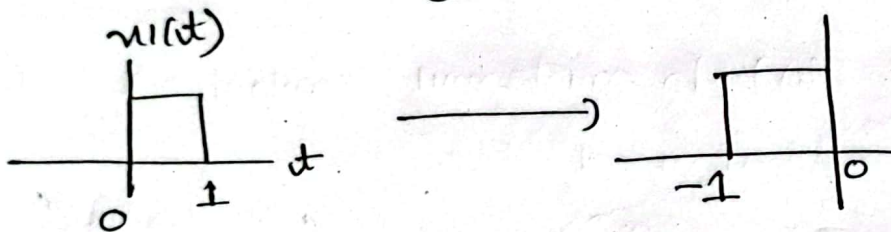
2nd time and more:

- i) shifting
- ii) finding Area.

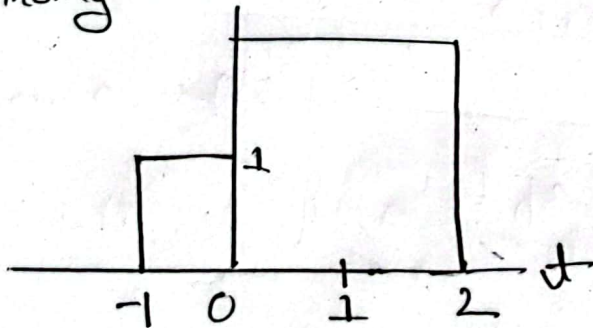
Q#107 find convolution integral $u_1(t) * u_2(t)$.



1st time: i) folding

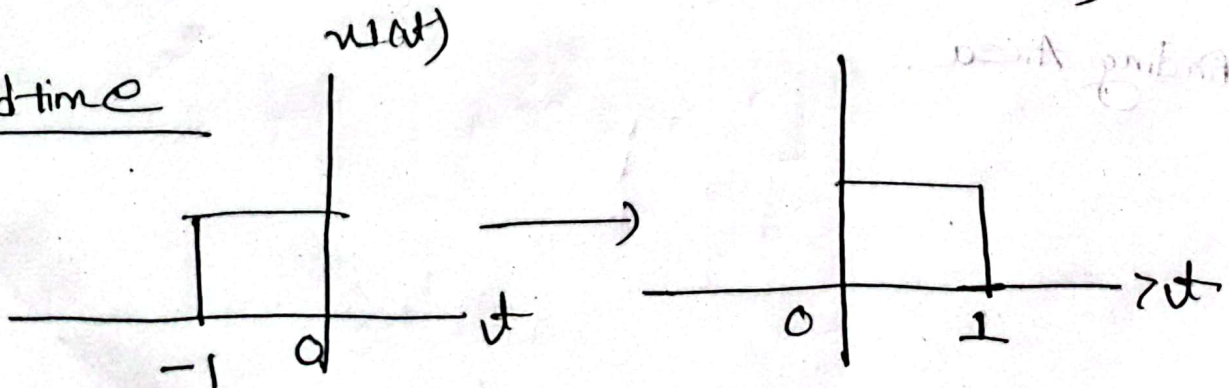


ii) finding Area

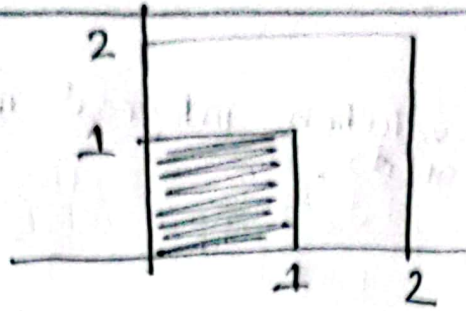


Area $u[0] = 0$.

2nd time

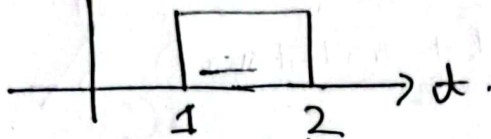


ii) finding area:

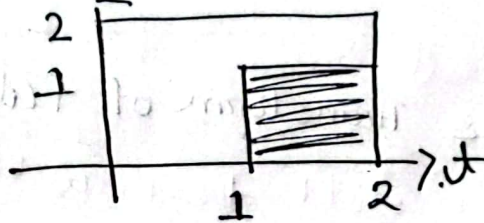


Area $n[1] = 1 \times 1 = 1$
 $n + (n-2)$

iii) Shifting:

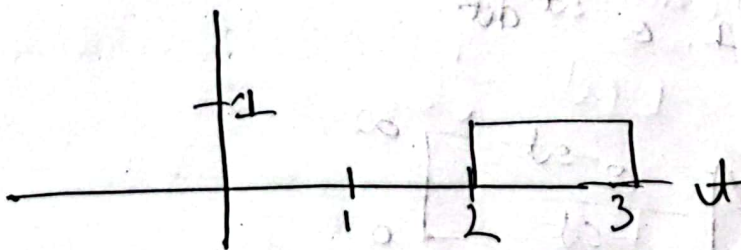


iv) finding area:

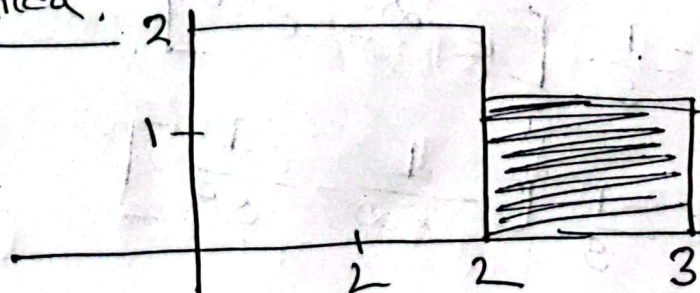


Area: $n[2] = 1 \times 1 = 1$

v) Shifting: $n + (x-3)$



finding Area:



Area = $n[3] = 0$

Istahadul Hoque Tasin

C221059, 4BM

hence the convolution integral is $x(t)$.

